

Peer Review File

Manuscript Title: Hallmarks of transcriptional intra-tumor heterogeneity across a thousand tumors

Reviewer Comments & Author Rebuttals

Reviewer Reports on the Initial Version:

Referee expertise:

Referee #1: single cell analyses including cancer, tumour microenvironment, computational expertise

Referee #2: single cell analyses of tumours, computational expertise

Referee #3: single cell analyses, plasticity

Referees' comments:

Referee #1 (Remarks to the Author):

Remarks to the Author

In their manuscript "The hallmarks of transcriptional intra-tumor heterogeneity across a thousand tumors", Gavish et al. curated, annotated and integrated single-cell RNA sequencing (scRNA-seq) data from 77 different studies covering 1,163 tumor samples from 24 tumor types. Based on this pan-cancer dataset and mainly focusing on the malignant cell population, the authors analyzed the transcriptional intra-tumor heterogeneity (ITH) in individual tumors, which is represented by expression programs/modules that consist of co-expressed genes and are up-regulated in subpopulations of cells in individual tumors. Further, by clustering intra-tumor expression programs, the authors reported 41 consensus meta-programs (MPs) that are shared by multiple tumors. Through functional annotations and comparison across tumor types, the authors identified that these MPs represent diverse cellular processes including those either shared by different tumor types (i.e., cell cycle and stress) or tumor type/lineage-specific (i.e., metal response). Then, the authors extended the MP analysis to the six most common non-malignant cell types in the tumor microenvironment (TME), compared non-malignant MPs to malignant ones, and utilized these to investigate cell-cell interactions within the TME. Finally, the authors have made the integrated pan-cancer dataset, annotated MPs and other analyses/results available on the Curated Cancer Cell Atlas (3CA) website and in the supplementary materials of the manuscript.

Overall, this is a well-designed study with high quality data and analyses. The endeavor of putting together a comprehensive pan-cancer compendium and dissecting transcriptional ITH across tumor types is impressive. The dataset and annotated meta-programs will be useful as resources for single-cell studies of various tumors and intra-tumor expression heterogeneity in the future. However, the concept of deciphering intra-tumor programs and meta-programs itself is not new and has been widely seen in many previous studies. The results revealed from the pan-cancer comparison are interesting, but most of them are not surprising. Besides, the manuscript needs more necessary

validations and is currently lacking the relevance to clinical implications that one would expect to see.

Major points

1. ITH is a central challenge of cancer therapeutics, which is also expected to associate with the survival rates of cancer patients. The authors should really leverage this pan-cancer dataset and the deciphered MPs to explore how MPs correlate with the prognosis/survival of patients with different cancers. Potentially this can be done by either scoring bulk RNA-seq data using MP gene lists (Supposedly, the authors can calculate malignant cell specific expression scores by subtracting TME contribution) or performing deconvolution. It could be interesting to see if a particular MP shows a different or even opposite correlation with prognosis in different cancer types. Similarly, it is not clear how different MPs associate with treatment failure and metastasis although these have been mentioned in the introduction part.
2. Although it has been mentioned by the manuscript that the identified MPs are less dependent on genetic features or genotypes of tumors. Can the authors find any associations between MPs in the current study and specific genomic alterations (i.e., driver mutations) in various cancer types? Based on current data (and meta data), can the authors find any other potential factors that drive MP expression in various cancer types?
3. Through comparison across cancer types, the authors found that some context-specific MPs had unexpected cancer type distributions. For example, MP30, which resembles the signature of the “classical” subtype of pancreatic adenocarcinoma (PDAC), was also identified in multiple non-PDAC cancer types. This finding seems interesting, but the underlying mechanism remains unclear. Can the authors delve deeper into this and perhaps try to infer the gene regulatory networks (GRN) based on scRNA-seq data? It will also be useful if the GRN inference analysis can be extended to other MPs in the pan-cancer dataset. Potentially, the results can be added to the Supplementary Note and the 3CA website, which will boost value of the manuscript as a resource.
4. One major comment I want to make here is: the manuscript needs more necessary validations (i.e., RNAScope and IHC staining) on the findings, such as MPs and associations between TME cell types in various cancer types. I do appreciate the NMF method has been proven robust in previous studies, but some experimentally validations are still necessary to corroborate findings such as (but not just) MP30 in non-PDAC cancers.
5. The authors conducted comparison between MPs in malignant cells and those in non-malignant epithelial cells. It is interesting to see that most MPs showed high similarities between the two cell types, which implies these common MPs are operative throughout the process of malignant transformation. However, for those malignant-specific or oncogenic MPs, it would be interesting to further look into the time points of their occurrence during tumorigenesis. Is there any data from pre-cancerous tissues that can be potentially integrated and analyzed? It could be interesting to compare MPs in normal epithelial cells, pre-cancerous epithelial cells and cancer cells. Similarly, the comparison between primary and metastatic tumors can also be of interest. Is there any data from metastatic tumors (lymph node or distant metastasis) can be integrated and analyzed? Or can the authors check MPs in bulk RNA-seq data from pre-cancerous tissues and metastatic tumors?
6. The authors inferred potential associations between cell types in TME by examining the co-occurrence between MPs of different cell types. Can the authors further validate this in larger cohorts by using bulk RNA-seq data (i.e., through deconvolution)?
7. In the current manuscript, common MPs (i.e., EMT programs) were found to be shared by multiple cancer types (Figure 3A) and cell types (Figure 4C). In addition to discussing the similarities, can the

authors also have more discussion regarding differences of same MPs in different cancer types and cell types?

Minor comments:

1. Can the authors further discuss global cell state and relative cell state, and how they influence tumor behaviors?
2. Are there any tumor types in the current study simultaneously having full-length data and 10X data? How do different data types influence the analysis?
3. The authors compared MPs in the manuscript to those previously identified from cancer cell lines. Can the authors provide a supplementary table to summarize the comparison?
4. In the data pre-processing, samples with unresolved CNA patterns were excluded from further analysis. Can the authors clarify how they defined unresolved CNA patterns?
5. Can the authors discuss and comment more on MHC II MP? How does the expression of MHC II on tumor cells potentially influence their behaviors?
6. On 3CA website, is there a place that one can query with new sets of genes and compare to the defined MPs?

Referee #2 (Remarks to the Author):

In this study, Gavish et al. collect single-cell RNA-seq datasets from 77 published studies covering 1,163 tumor samples from 24 tumor types to establish an online database. The authors collect published cell type annotations and combine this with de novo annotation and copy-number alteration analysis for calling malignant cells to provide a well-curated and extendable database. They use this resource to explore intra-tumor heterogeneity (ITH) by inferring meta-programs (MPs) of gene expression that co-vary within malignant cells of specific tumor types or across tumor types. This approach is then also extended to the tumor micro-environment, in order to search for shared meta-programs with malignant cells as well as co-varying meta-programs.

This is a useful resource that will facilitate meta-analyses across large cohorts and different tumor types and provides additional information and curation compared to the original studies. However, the authors do not sufficiently demonstrate that the ITH analysis they perform can lead to genuinely novel and interesting insights. The scope of the MP approach remains limited and many of the observations are not unexpected (such as common MPs between malignant and non-malignant epithelial cells) or analyzed too superficially (see detailed comments). Thus, in its current form I do not see this resource as a major advance for the cancer field.

Specific comments:

1. The online atlas could be a useful data collection tool for the community. However, I was unable to find the metadata (cell type, sample annotation, and CNA per cell) as a text file that can be used for data analysis. The authors merely provide summary plots. UMAPs highlighting the sample IDs should also be added.
2. The authors report many already known meta-programs associated with tumor cell type

variability. There are interesting and important potential programs that were not explicitly reported. In particular, I was curious whether the authors recovered programs associated with immune-suppression, or with the metabolic state of the tumor cells. Those classifications could be important for stratification of tumor types.

3. The Jaccard similarity computed for the inference of consensus metaprograms is sensitive to the input feature set for each sample. If one sample exhibits substantially lower sensitivity compared to another, sets of genes with high Jaccard similarity may be dis-favoured. This could lead to a counterselection of MPs comprising lowly expressed genes, consistent with the finding that the most prominent MPs comprise highly expressed genes (such as proliferation). Did the authors account for global differences in sensitivity across datasets, e.g., by only keeping the intersect of expressed genes across samples?

4. Please show to what extent cell type clusters correspond to absence or presence of CNA calls. This would show if clusters are uniquely assigned to malignant and non-malignant cells and could address the question whether overlapping meta-programs are due to incorrect annotations of a fraction of cells within the same cluster, e.g., due to sparse coverage or other technical noise. In particular, could the correlation of MYC expression and proliferation MPs between malignant and non-malignant epithelial TME be explained by incorrect annotations of malignant and non-malignant cells?

5. The authors consider the fraction of tumours of a given cancer type for which a robust NMF is assigned to a particular MP as a proxy for the variability of this MP. This interpretation is misleading, since robustness of NMF indicates significant covariation of a group of genes contributing to a factor without implying high variability of a factor per se. If the authors want to make this point they should explicitly show this by a suitable method, such as PCA or fano factor.

6. The authors highlight shared MPs between malignant and non-malignant cells. They make some surprising observations such as shared cell cycle and MYC MPs. It is unclear, however, what that really means. Do the authors want to make the claim that MYC or proliferation is variable in non-malignant cells or indeed up-regulated. I am not aware of induction of proliferation in non-malignant cells by tumor cells - perhaps with the exception of immune cell activation - but the data presented in Figure 5 suggest that this is the case for all populations. This should be analyzed in more detail. Would a similar finding be obtained by differential gene expression analysis? The authors should also validate this in situ by immune fluorescence analysis of proliferation or MYC genes together with cancer cell markers. This would strongly support the usefulness of the atlas and the analysis strategy.

7. The MP analysis reveals some interesting findings, but remains superficial. For instance, the authors report a correlation of proliferation of CD4+ and CD8+ T cells, but not their abundance, with macrophage abundance. How could this be explained? Naïvely, one would assume that increased proliferation leads to higher abundance, unless T cell are actively killed or migrate out of the tumor tissue. Is there evidence for any of these processes in the MP analysis?

8. I was surprised that the authors did not observe an association between CD8+ T cell cytotoxicity and malignant cancer cell abundance or proliferation MPs? I would expect a negative correlation.

Along the same lines I was wondering whether any MPs related to T cell exhaustion were observed and how they would associate with malignant cell MPs? Is there any correlation between regulatory T cell abundance and malignant MPs? This resource could potentially be very informative for studying immune-tumour crosstalk, but this aspect remained under-investigated.

9. Another important aspect is the capacity of a tumor to metastasize? Do the studies collected here provided information on this aspect? It would be interesting to extract MPs that are specifically associated with the propensity to metastasize.

Referee #3 (Remarks to the Author):

In this manuscript, the authors examine cellular programs underlying intratumor heterogeneity with scRNA-seq data across a large number of studies and over a thousand tumor samples. Their analysis focuses on identifying systematic variable gene expression programs that are conserved across samples. They define 41 consensus “metaprograms” that arise in subpopulations within these tumor samples and provide further classification of these metaprograms into 11 hallmarks of intra tumor heterogeneity. They extend this system to the non-malignant cells in the dataset and use correlations between metaprograms across cell types to infer interactions. Ultimately, this paper outlines a broad description of intratumoral heterogeneity, which I believe will have a pretty significant impact on the field, specifically on how people classify and think about cellular heterogeneity in tumors.

This work builds upon previous work from this group and others describing conserved programs of tumor heterogeneity that exist across different cancer types. The scale of this work is much larger and incorporates a truly impressive number of different data sets. Further they are careful with issues of data integration and batch correction, instead they do not integrate the data, but rather measure the pattern of variability within each tumor.

While the discussion outlines some of the potential reasons that this study may have missed additional hallmarks of intratumoral heterogeneity, I think they likely do capture the largest categories given how many samples these are conserved across. While other studies or groups may attribute different names or classification systems, I generally think that their approach captures most of the variation (at least the more common variation) across tumors.

I have a few minor suggestions and clarifications for the authors:

In looking at the datasets through the browser, I noticed that many come from a variety of different chemistries. Are there biases or issues arising with the different chemistries used? Are there programs that are only detected from a particular technology platform? Does sequencing chemistry correlate with the ability to recover specific metaprograms? I am wondering if there are considerations related to lowly expressed genes or even genes that are more difficult to sequence for any reason.

One question about the web browser, are there compiled datasets available for download or tools to analyze the data across all the samples? Is this intentional and related to the fact that the data is not integrated? From the website, it is clear how to download individual datasets used in this paper, but I think it would be useful to the community to be able to access the compiled data or some form of compiled analysis results (I believe this is what this study really adds in terms of data).

How were the cut-offs selected for grouping the different metaprograms into the 11 hallmarks? I see the clustering in 2A, but how were the boundaries defined for including each metaprogram into a hallmark category?

Author Rebuttals to Initial Comments:

Dear reviewers,

We thank you for the constructive feedback. Over the last 10 months, we have worked to comprehensively address these comments through various additional analyses, experiments and modifications in the text. We believe that the revisions have advanced our study significantly, and that our resulting manuscript will be an important contribution for the cancer research community. Please find below a point-by-point response to all comments.

Major additions to the revised manuscript include:

- (i) **Experimental validations:** We validated multiple observations described in the manuscript using extensive immunohistochemistry (IHC) staining of six cancer types, as well as by re-analysis of published spatial proteomics (CODEX) and transcriptomics (VISIUM) datasets.
- (ii) **Associations of meta-programs (MP) with clinical and genetic features:** We evaluated the expression of MPs in TCGA bulk RNA-seq data and describe significant associations with various clinical features, with mutations and with copy number alterations (CNA).
- (iii) **Evaluating the link between MPs and genetic subclones:** We systematically identified genetic subclones throughout the scRNA-seq compendium, based on inference of CNAs. This allowed us to determine the degree to which subclones may account for variability in MP expression.
- (iv) **Inference of MP regulators:** We applied SCENIC to the entire scRNA-seq compendium and predicted transcription regulators for most MPs.
- (v) **Comparison of MPs between tumors and non-tumor tissues:** We derived MPs from extensive datasets of non-tumor tissues and compared them to their malignant counterparts.
- (vi) **Updating the website with new analysis and functionalities,** enhancing its utility as a resource.
- (vii) **New results:** the analyses and experiments noted above led to new results such as the context-dependent coupling of MHC-II with interferon responses and the PDAC-classical MP in HNSCC.

As already discussed with the editor, during these revisions a relevant study demonstrating similarity of cancer cell states between several cancer types was published in *Nature Genetics* (<https://www.nature.com/articles/s41588-022-01141-9>). In the revised manuscript we cite this work. However, we note the different scope of our study:

- (1) **Cancer state analysis** – 1,163 cancer samples included in our study vs. 62 samples in the other study, allowing us to determine many more cell states (41 vs. 16), including lineage-specific states, and to robustly define the consensus programs and their distribution across diverse cancer types.
- (2) **TME cell states** – in addition to cell states of cancer cells, we also defined cell states in the six common TME cell types and further uncovered their interactions with cancer cell states.
- (3) **Resource and its utility** – the 3CA website provides a comprehensive curated database to which we are committed to continue adding data, analyses and functionality.

Aside from these three major points, various types of analyses that we performed were not included in the other study. We therefore believe that, despite certain similarities, the new work by Yanai's group would not diminish from the significance of our work, as our analyses of cell states across malignant cells are by far more comprehensive; the definition of states for non-cancer TME cells, the associated interactions across cell types and many other analyses are unique to our work; and our resulting dataset and website will be highly useful as a foundational resource for the cancer research community.

Reviewer #1

In their manuscript “The hallmarks of transcriptional intra-tumor heterogeneity across a thousand tumors”, Gavish et al. curated, annotated and integrated single-cell RNA sequencing (scRNA-seq) data from 77 different studies covering 1,163 tumor samples from 24 tumor types. Based on this pan-cancer dataset and mainly focusing on the malignant cell population, the authors analyzed the transcriptional intra-tumor heterogeneity (ITH) in individual tumors, which is represented by expression programs/modules that consist of co-expressed genes and are up-regulated in subpopulations of cells in individual tumors. Further, by clustering intra-tumor expression programs, the authors reported 41 consensus meta-programs (MPs) that are shared by multiple tumors. Through functional annotations and comparison across tumor types, the authors identified that these MPs represent diverse cellular processes including those either shared by different tumor types (i.e., cell cycle and stress) or tumor type/lineage-specific (i.e., metal response). Then, the authors extended the MP analysis to the six most common non-malignant cell types in the tumor microenvironment (TME), compared non-malignant MPs to malignant ones, and utilized these to investigate cell-cell interactions within the TME. Finally, the authors have made the integrated pan-cancer dataset, annotated MPs and other analyses/results available on the Curated Cancer Cell Atlas (3CA) website and in the supplementary materials of the manuscript.

Overall, this is a well-designed study with high quality data and analyses. The endeavor of putting together a comprehensive pan-cancer compendium and dissecting transcriptional ITH across tumor types is impressive. The dataset and annotated meta-programs will be useful as resources for single-cell studies of various tumors and intra-tumor expression heterogeneity in the future. However, the concept of deciphering intra-tumor programs and meta-programs itself is not new and has been widely seen in many previous studies. The results revealed from the pan-cancer comparison are interesting, but most of them are not surprising. Besides, the manuscript needs more necessary validations and is currently lacking the relevance to clinical implications that one would expect to see.

Major point 1

“ITH is a central challenge of cancer therapeutics, which is also expected to associate with the survival rates of cancer patients. The authors should really leverage this pan-cancer dataset and the deciphered MPs to explore how MPs correlate with the prognosis/survival of patients with different cancers. Potentially this can be done by either scoring bulk RNA-seq data using MP gene lists (Supposedly, the authors can calculate malignant cell specific expression scores by subtracting TME contribution) or performing deconvolution. It could be interesting to see if a particular MP shows a different or even opposite correlation with prognosis in different cancer types. Similarly, it is not clear how different MPs associate with treatment failure and metastasis although these have been mentioned in the introduction part.”

We fully agree with the reviewer on the importance of linking programs of ITH to clinical features, and on the value of integrating our results with large, well-annotated bulk RNA-seq cohorts, especially given the sparsity of clinical annotations in scRNA-seq cohorts. To this end, as suggested by the reviewer, we computed scores for each malignant-cell MP in bulk RNA-seq data from TCGA

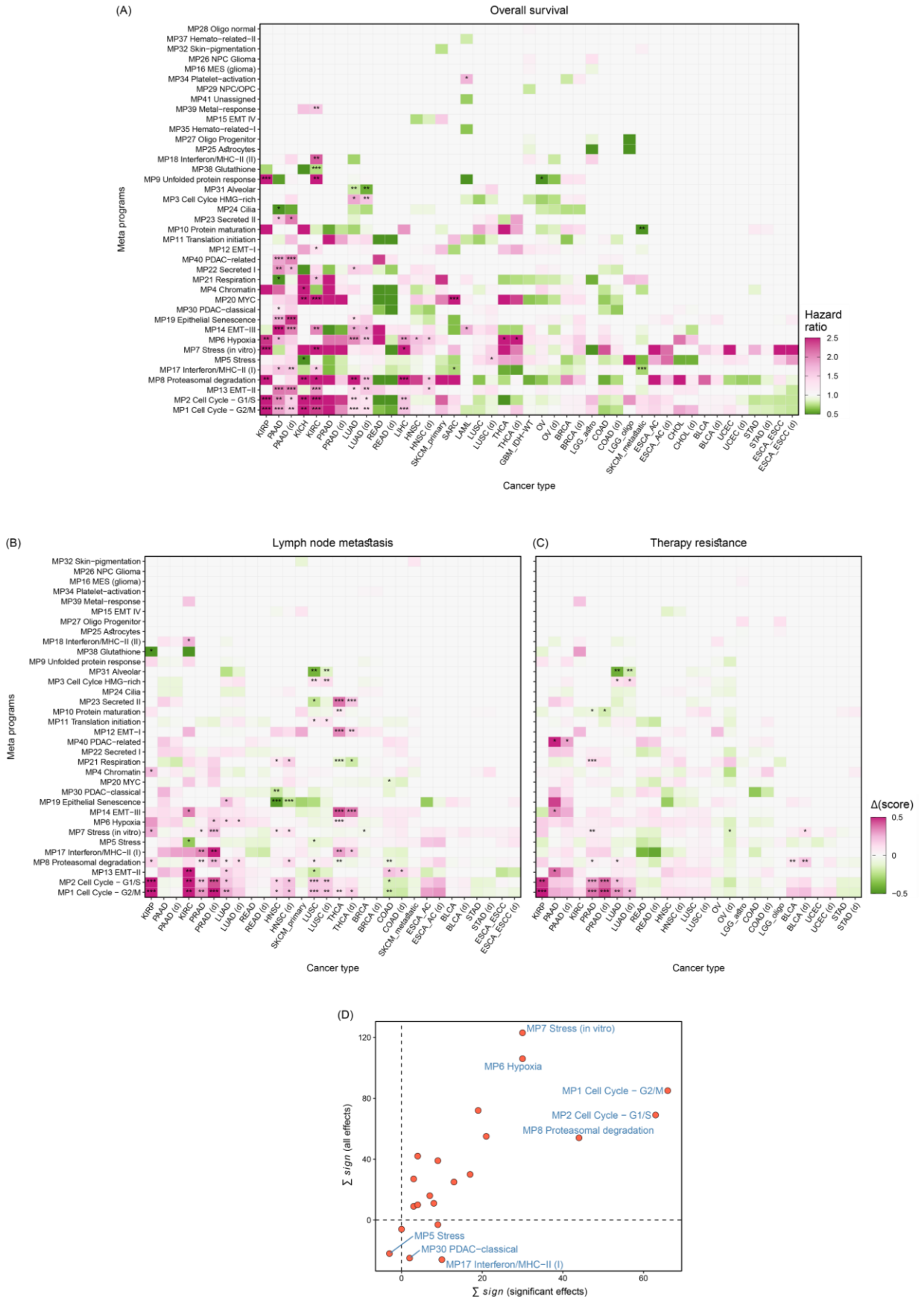
for those cancer types in which this MP was detected by scRNA-seq. MP scores were defined both with and without deconvolution (see below). We then assessed the associations between these MP scores and a variety of clinical features, including overall survival (**Response Figure 1A**), progression-free survival, grade, AJCC stage, T stage, M stage, lymph node metastasis (**Response Figure 1B**) and therapy resistance (**Response Figure 1C**) (see **Methods** for additional details).

Our results reveal many significant associations (see new **Table S8** for a full list). Some associations are unique to a specific cancer type, such as the associations of the Alveolar MP with higher overall survival and with decreased therapy resistance in lung adenocarcinoma (LUAD). Similarly, the Glutathione MP was associated with higher overall survival only in Kidney clear cell carcinoma (KIRC), an association which we investigated further in the context of genomic alterations [see next section].

In contrast, other associations are more consistent across cancer types (**Response Figure 1D**). The MPs most consistently correlated with worse outcomes (across all examined clinical features) include cell cycle, hypoxia and proteasomal degradation, along with a stress MP seen predominantly in cell lines. Associations with positive outcomes were less consistent across cancers.

There are two important caveats to this analysis. Firstly, as discussed in the manuscript, MPs may be expressed variably or uniformly in a given tumor, and while our scRNA-seq analysis identified programs by their variability, any analysis in bulk RNA-seq can detect only their average expression within a tumor. Secondly, as the reviewer indicated, the TME may contribute to the overall signal for a given MP. To control for the TME contribution, we repeated this analysis with data for inferred cancer cell profiles obtained via the deconvolution algorithm CIBERSORTx. This data yielded similar results to the non-deconvolved data (**Response Figure 1A-C**), indicating that either the TME has little confounding effect or deconvolution methods are not sufficient to fully control for TME confounders.

This analysis has been added to the manuscript in **page 7, Figure S8 and Table S8**.



Response Figure 1. Associations of MPs with clinical features in TCGA data. (A) Heatmap showing the hazard ratio of scores for MPs (rows) with respect to overall survival in each cancer type (columns). Purple and green represent association with worse and better survival, respectively. For relevant carcinoma types, the columns of results from data deconvolved by CIBERSORTx are presented with the suffix “(d)”, adjacent to the corresponding column for non-deconvolved data. Significant associations are labelled with ‘*’ for $p < 0.05$, ‘**’ for $p < 0.01$ and ‘***’ for $p < 0.001$. Significance levels were computed via Cox regression and adjusted by Benjamini-Hochberg correction. **(B-C)** Heatmaps as in (A) showing the difference in average scores between tumors stratified by lymph node metastasis and therapy resistance, respectively. Significance levels were computed by t-test and adjusted by Benjamini-Hochberg correction. **(D)** Scatterplot showing consistency of associations of MPs with clinical features in TCGA data. Each point represents an MP, its X-value being the sum of the signs (1 for positive effect, -1 for negative effect) of all significant effects (adjusted p value < 0.05) for that MP across all cancer types and clinical features, and its Y-value being the sum of the signs of all effects for that MP. Selected points are labelled with their corresponding MPs.

Major point 2

“Although it has been mentioned by the manuscript that the identified MPs are less dependent on genetic features or genotypes of tumors. Can the authors find any associations between MPs in the current study and specific genomic alterations (i.e., driver mutations) in various cancer types? Based on current data (and meta data), can the authors find any other potential factors that drive MP expression in various cancer types?”

To address this important point, we performed two distinct types of analysis that are further described below:

- (1) We tested to what degree genetic subclones may explain diversity in MP expression within tumors, using our scRNA-seq compendium.
- (2) We searched for mutations and CNAs that may explain diversity in MP expression between tumors, using TCGA bulk RNA-seq data.

Single cell analysis of genetic subclones

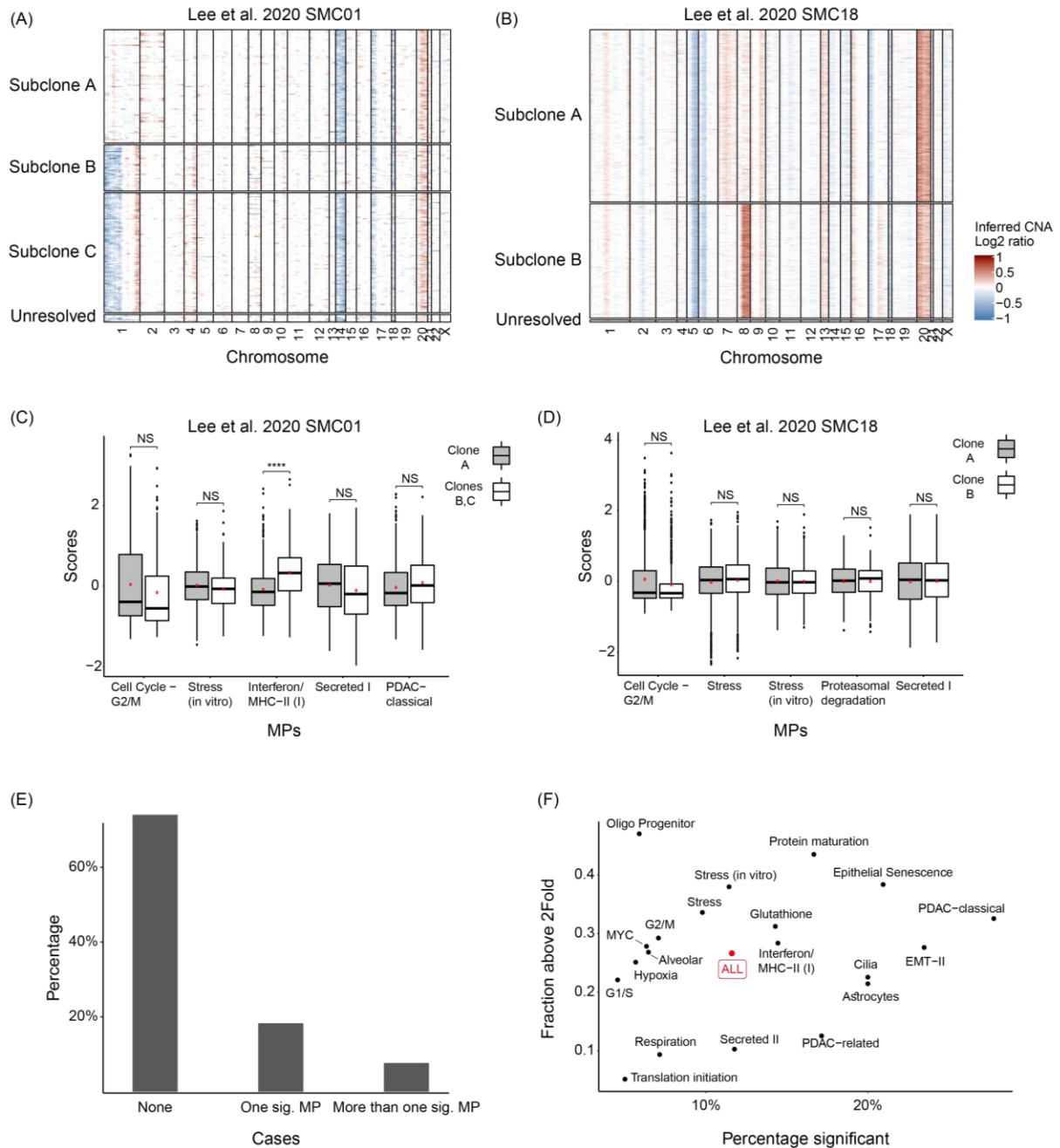
As previously described, we systematically inferred CNA patterns throughout the scRNA-seq compendium in order to identify the malignant cells. In the revision, we utilized the inferred CNA profiles for further analysis. First, we divided the malignant cells in each tumor to genetic subclones, using a strict approach (retaining only high-confidence subclonal differences). We were able to robustly define multiple subclones within 16% of the samples, typically identifying 2-3 subclones per tumor that differ by 1-2 gains or losses of complete chromosome or chromosome arms. Second, we examined the differences in MP expression across the subclones within each of those tumors. When such differences are significant we infer that genetic differences *could* be the driver of MP expression differences. We then estimated the fraction of genetic subclones that are associated with significant MP expression differences.

Defining subclones: We clustered the inferred CNA matrix of each sample using Louvain clustering. To ensure that clusters correspond to distinct genetic subclones, we then iteratively merged all pairs of clusters for which we could not find any chromosome arm with evidence for a distinct copy number (see Methods for exact definitions). After merging, the remaining clusters were considered as distinct subclones and were examined for differences in MP expression. **Response Figure 2A-B** shows two examples from a colorectal scRNA-seq dataset (Lee et al. 2020). Sample SMC01 (panel A) harbors three subclones and sample SMC18 (panel B) harbors two subclones.

Defining differential expression (DE) between subclones: For each subclone, we compared its MP expression to that of all other subclones in the same tumor and examined whether there is significant DE based on a two-sample t-test across cells (with p-value < 0.05) and a fold-change above 1.5. For example, in sample SMC01, only Interferon/MHC-II (I) was DE in one subclone (compared to the other two), while in sample SMC18 none of the MPs were DE between subclones (**Response Figure 2C-D**).

Overall, we found that 76% of the identified subclones did not have significant associations with any of the MPs detected in the corresponding tumor (as in the example of SMC18), 17% of the subclones had a significant associations with only one MP (as in the example of SMC01) and 7% had a significant association with more than one MP (**Response Figure 2E**). Moreover, even in those relatively few subclones with significant associations, only a limited fraction of the cells in the subclones had higher (or lower) expression of the MP by at least 2-fold (**Response Figure 2F**). This fraction varied between subclones and MPs but, on average, was approximately a quarter. We note that similar results were obtained with alternative parameters (data not shown).

Taken together, the consistency between genetic subclones (as defined by CNAs) and MP heterogeneity is quite limited and suggests a major contribution of non-genetic plasticity. In order to identify *specific* associations between CNAs and MPs we turn to the next analysis of bulk samples, in which we are better powered to identify such associations.



Response Figure 2. Genetic subclones explain limited variability in MP expression. (A-B) CNA patterns for two samples from the Lee et al. 2020 colorectal study, in which 3 and 2 subclones were detected, respectively. Few cells were unresolved. **(C-D)** Boxplots depicting MP scores across cells in subclones detected in (A) and (B), for all MPs detected in the respective samples. Also shown is the significance of the difference between subclones. In sample SMC01, only the Interferon/MHC-II (I) was significantly different between subclone A and the other two subclones; in sample SMC18, none of the MPs was significantly different between subclones. **(E)** Percentage of subclones with zero, one or >1 significant MPs in comparison to the other subclones from the same tumor. **(F)** For each MP, the X-axis shows the percentage of subclones in which that MP was significantly different from other subclones in the same tumor (calculated only among subclones in which the MP is detected); for that subset of significant subclones, the Y-axis shows the mean

fraction of cells with at least a 2-fold difference from the average of the tumor: i.e. cells with score above 1 for subclones with increased MP expression, or cells with score below -1 for subclones with decreased MP expression. Thus, even in subclones with a significant upregulation/downregulation, only a minority of cells have larger than 2-fold effect. Red dot ('ALL') indicates the average across all MPs.

Bulk analysis of mutations and CNAs

Our scRNA-seq analysis has limited power to detect associations between MPs and genetics, due to the number of profiled samples and the lack of genetic information at the single cell level. While we can infer CNAs, our ability to detect mutations from scRNA-seq data is highly limited (data not shown). Thus, an alternative strategy to identify genetic associations with MPs is to leverage highly annotated bulk samples. Since TCGA samples are genetically well annotated, we can search for tumor mutations (and CNAs) that correlate with a high (or low) average (i.e. bulk) expression of MPs.

We examined the associations between our computed bulk MP scores and mutations in TCGA data for the relevant cancer types. More precisely, we considered a number of genomic events including common cancer driver mutations and focal or whole-chromosome-arm gains/losses, comparing MP scores between tumors with and without each such event. As shown in **Response Figure 3** we identified a large number of significant associations (see **Table S7** for a full list). The strongest associations include high cell cycle expression in tumors with RB1 and TP53 mutations, with the latter having the highest overall number of significant associations. Additional selected associations are described further below.

Interferon-response associations: Multiple associations were observed with the Interferon/MHC-II (I) program. Interestingly, in head and neck cancer (HNSCC), this MP was strongly correlated with the presence of mutations in CASP8 (**Response Figure 3A,E**). CASP8 is involved in apoptosis and its expression is induced by interferon (ref. 17). Thus, inactivating CASP8 mutations may enable cancer cells to survive in an interferon-high environment without activating apoptosis but while still inducing an interferon response program. This association remains strongly significant for the tumors having both CASP8 and HRAS mutations and lacking TP53 mutations (**Response Figure 3E**), a potential subpopulation also discussed in the TCGA study of HNSCC. In lung adenocarcinoma (LUAD), we observed strong negative associations of the Interferon/MHC-II (I) MP with mutations in STK11 and KEAP1, as well as a borderline significant association with mutations in NFE2L2 (NRF2), a transcription factor which is degraded by KEAP1 (**Response Figure 3A,F**).

PDAC-classical associations: In the previous version of our manuscript, following the observation of the PDAC-classical MP in several non-PDAC cancer types, we showed that this MP correlates in LUAD with KRAS mutations and a mucinous subtype. Our latest genetic analysis further extends these results, revealing also a correlation between the PDAC-classical MP in HNSCC and mutations in the histone methyltransferase NSD1 (**Response Figure 3A,G**). The PDAC-classical MP is not significantly associated with previously defined HNSCC subtypes, though it is significantly enriched in laryngeal tumors ($p=8.2 \times 10^{-8}$). Nevertheless, the association of the PDAC-classical MP with

NSD1 mutations remains significant even after separating HNSCCs to laryngeal and non-laryngeal tumors (**Response Figure 3G**).

To further understand the link between NSD1-mutant HNSCC and the tumors scoring highly for the MP in PDAC and in LUAD, we defined a PDAC-specific, a LUAD-specific and a shared signature of the high-scoring tumors. The shared signature and the PDAC-specific signature were upregulated in NSD1-mutant HNSCCs, while the LUAD-specific signature was not upregulated (**Response Figure 3H**). Thus, a consistent signature, corresponding to a subset of the PDAC-classical MP genes, defines related subtypes of three cancer types, while additional genes are specific to one or two of these subtypes.

Finally, our analysis revealed a strong association between the Glutathione MP and various copy number aberrations in kidney clear cell carcinoma (KIRC) (**Response Figure 3A-D**). The distribution of Glutathione scores in KIRC reveals an extended negative tail (**Response Figure 3I**), suggestive of a distinct glutathione-low subpopulation which typically lacks the canonical VHL and PBRM1 mutations, while being modestly enriched with focal deletions of CDKN2A (**Response Figure 3J**). This KIRC subpopulation has not, to the best of our knowledge, been described before, but may be clinically important as it correlates strongly with poor survival and is significantly more proliferative than the majority of KIRC tumors (**Response Figure 3K-L**).

Taken together, we performed extensive analyses of the link between genetics and MPs, both at the single cell level and across TCGA tumors. We find that only a minority of intra-tumor heterogeneity in MP expression may be linked to CNA-based genetic subclones, but identify a large number of associations between tumor genetics and MP expression, including several cases that suggests potential tumor subtypes that are linked to MP expression.

These results are described in the manuscript in **page 6, Figure 3D, Figure S5-S6 and Table S7**.

Response Figure 3. Associations of MPs with genomic alterations in TCGA data. (A-B) Heatmaps showing significant associations between MP scores (columns) and mutations or focal amplifications/deletions (rows), colored by the cancer types in which they were observed, or in red if observed in multiple cancer types. Panels separate positive and negative effects, that is, associations where the difference in score between tumors with and without a given genomic alteration is positive or negative, respectively. Significance levels were computed by t-test and adjusted by Benjamini-Hochberg correction. **(C-D)** Same as A-B for chromosome arm gains and losses. **(E)** Boxplot showing scores (Y-axis) for the Interferon/MHC-II (I) meta-program in HNSCC tumors (points), comparing CASP8-mut tumors, or CASP8-mut/HRAS-mut/TP53-wt tumors, with all others (X-axis). Significance levels were computed by t-test. **(F)** Boxplot showing scores (Y-axis) for the Interferon/MHC-II (I) meta-program in LUAD tumors (points), comparing tumors with and without STK11 and KEAP1 mutations, respectively. Significance levels were computed by t-test. **(G)** Boxplot showing scores (Y-axis) for the PDAC-classical meta-program in NSD1-mut and NSD1-wt HNSCC tumors (points), first amongst all HNSCC tumors, then separately for non-laryngeal and laryngeal tumors. Significance levels were computed by t-test. **(H)** Boxplot showing scores (Y-axis) in NSD1-mut and NSD1-wt HNSC tumors (points) for three MP30-related gene-sets: those specific to the Mucinous LUAD tumors, those specific to Classical PDAC tumors, and those shared between Mucinous LUAD and Classical PDAC tumors. Significance levels were computed by t test. **(I)** Histogram of scores for the Glutathione meta-program in KIRC tumors. The dashed red line indicates the chosen threshold of -1, used to define Glutathione-low and -high populations. **(J)** Bar plot showing the percentage of KIRC tumors (Y-axis) in Glutathione-low and -high categories (color) having alterations in VHL, PBRM1 and CDKN2A genes (X-axis). Significance levels were computed by hypergeometric test, with the notation ‘*’, ‘**’ and ‘***’ indicating $p < 0.05$, $p < 0.01$ and $p < 0.001$, respectively. **(K)** Kaplan-Meier plot comparing survival probability (Y-axis) across time (X-axis) between KIRC tumors in Glutathione-low and -high categories (color), with P-value computed by log-rank test. **(L)** Boxplot comparing the average scores for the G1/S and G2/M MPs (Y-axis) between KIRC tumors (points) in the Glutathione-low and -high categories (color), with P-value computed by t-test.

Major point 3

“Through comparison across cancer types, the authors found that some context-specific MPs had unexpected cancer type distributions. For example, MP30, which resembles the signature of the “classical” subtype of pancreatic adenocarcinoma (PDAC), was also identified in multiple non-PDAC cancer types. This finding seems interesting, but the underlying mechanism remains unclear. Can the authors delve deeper into this and perhaps try to infer the gene regulatory networks (GRN) based on scRNA-seq data? It will also be useful if the GRN inference analysis can be extended to other MPs in the pan-cancer dataset. Potentially, the results can be added to the Supplementary Note and the 3CA website, which will boost value of the manuscript as a resource.”

We thank the reviewer for this constructive comment and fully agree that gene regulatory network inference is an important analysis that was not included in our previous manuscript. To address this, we applied the SCENIC method to each of the tumor samples in which we identified cancer MPs (1,012 scRNAseq tumors in total). In brief, the SCENIC workflow includes 3 main steps which were applied separately to each sample:

- (1) Sets of genes that are co-expressed with transcription factors (TFs) are identified by GENIE3.
- (2) Each co-expression module is subjected to cis-regulatory motif analysis using the RcisTarget algorithm. Modules with significant motif enrichment of the correct upstream regulator are retained, and they are pruned to remove indirect targets lacking motif support. These processed modules are referred to as regulons.
- (3) The AUCell algorithm scores the expression of each regulon in each cell.

Below, we first describe our analysis of MP30 (PDAC-classical) potential regulators, as requested by the reviewer, followed by pan-cancer analysis of regulators of all other MPs.

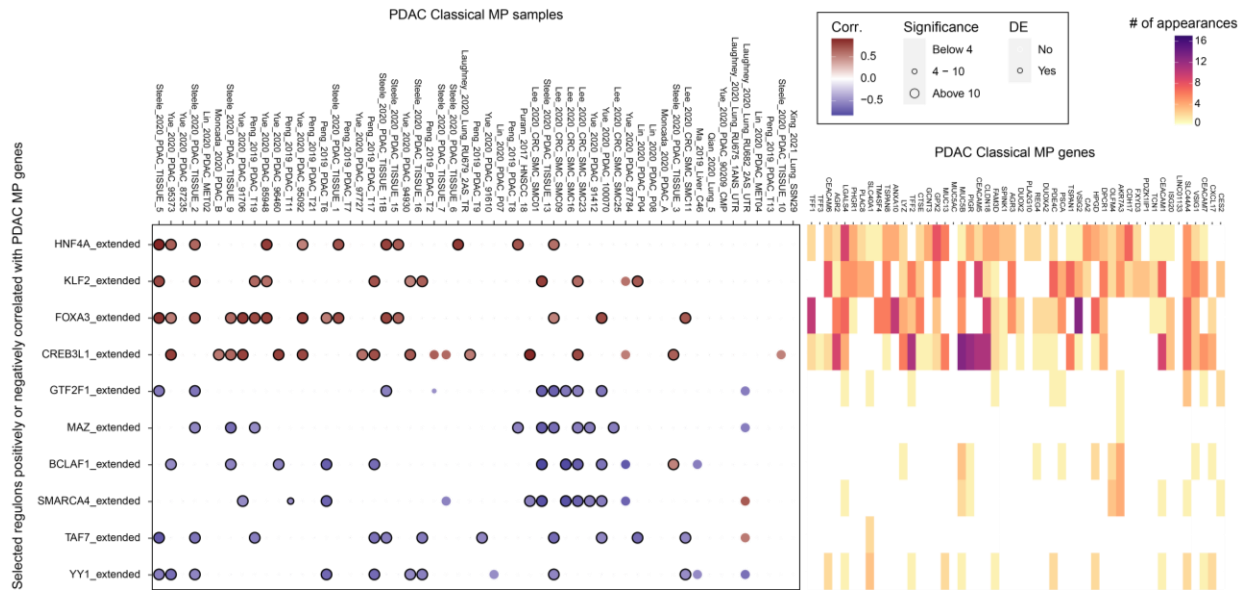
Prediction of regulators for MP30

For each of the samples that participated in MP30 (i.e. contributed a NMF program that was part of the MP), we first calculated the correlation between the cells' MP scores and their scores for all regulons defined by SCENIC. We then selected the regulons with a statistically significant correlation ($P < 0.05$ after FDR correction) that is above 0.5 or below -0.5. We retained the top 3 positively or negatively correlated regulons as potential MP regulators. We also retained regulons that were consistently selected in many samples (at least 10 different samples or at least 30% of all samples with the MP), whereas those selected only in one or very few samples were excluded.

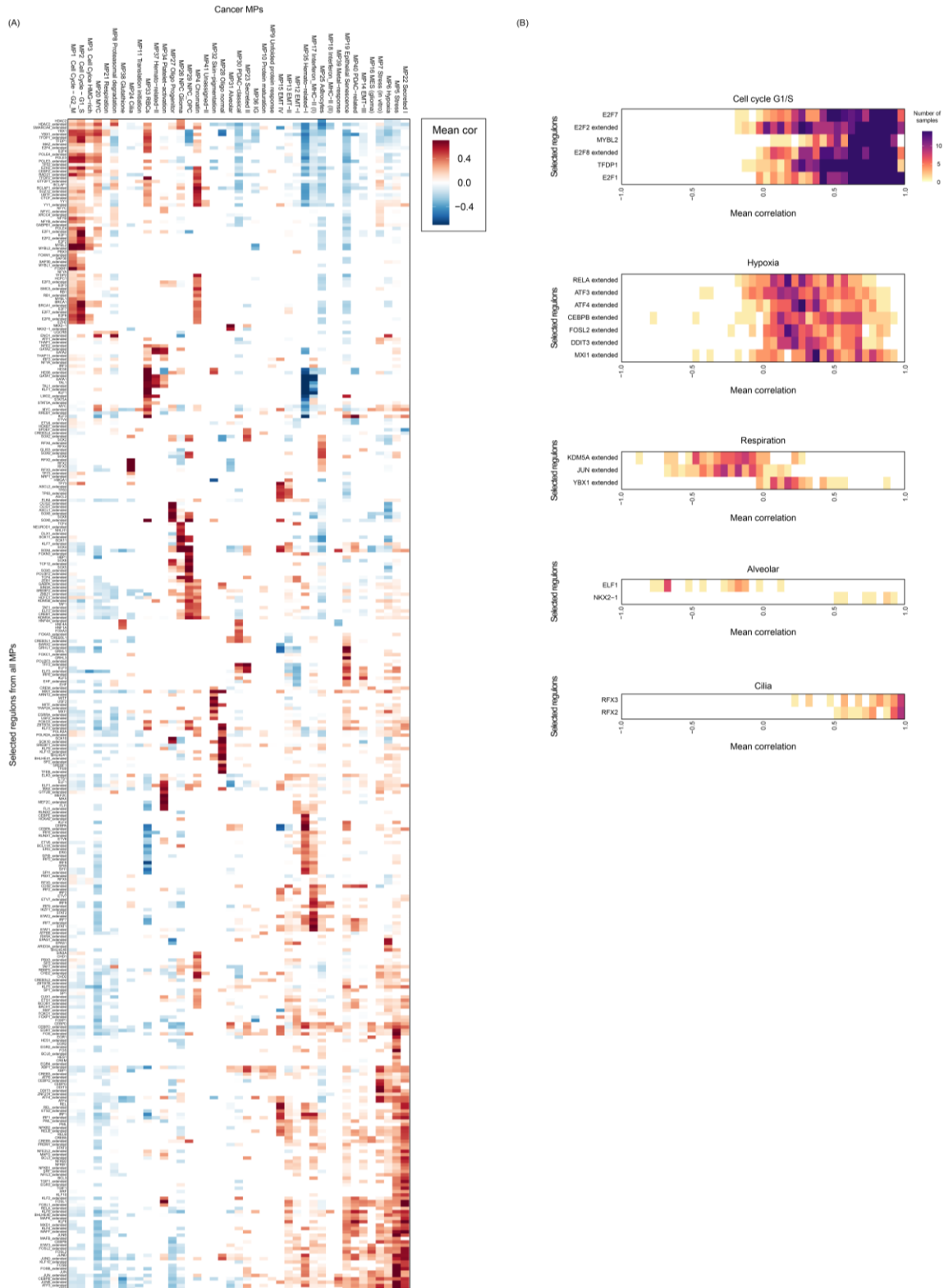
Response Figure 4 shows the regulons that were consistently positively correlated or negatively correlated with the PDAC-classical MP, and the inclusion of PDAC-classical MP genes in these regulons. Positively associated regulons include FOXA3 and CREB3L1. Notably, both of these have previously been shown to regulate the differentiation of Goblet cells which are specialized for mucin production. These results, along with the presence of multiple mucins (MUC13, MUC5AC, MUC5B) and multiple mucin-related genes (e.g. GCNT3 and TFF1-3) in MP30, and the association with *mucinous* lung adenocarcinoma, all lead us to suspect that MP30 reflects an aberrant (and possibly non-productive) activation of a mucin production program that could occur in multiple cancer types but is particularly common in PDAC. We describe this possibility in the revised manuscript at page 8, lines 22-28.

Prediction of regulators for all other MPs

Similar analysis was performed for all other MPs. Predicted MP regulators are provided in **Table S6** and in **Response Figure 5A**, which shows the mean correlations (across samples) between selected regulons and MP scores. The distribution of correlations across samples is shown for selected regulons in **Response Figure 5B** and additional examples are shown in **Fig. 2E**. These results are described in the manuscript in **page 6, Figure 2D-E, Figure S4** and **Table S6**. We also added the predicted regulators to the description of the individual MPs in **Supplementary Note 1** and in the future we will also add it to the **3CA website**.



Response Figure 4. Shared regulatory networks in the PDAC MP. The left panel shows the selected regulons that were positively or negatively correlated with the MP scores (had a respective correlation higher than 0.5 or lower than -0.5). Only cases with a significance score ($-\log_{10}(\text{p-value})$) higher than 4 after FDR correction are shown. Circle size indicates whether the significance was in the range of 4-10 or above 10, and cases in which the regulon scores were statistically different in sample cells with a high PDAC score (above 1) have a black circumference. The right panel shows the number of times each of the MP genes appeared in the corresponding regulons to the left that passed the correlation threshold.



Response Figure 5. MP regulators inferred by SCENIC. (A) Heatmap showing the mean correlations (across samples) between regulon scores and MP scores, for selected regulons. **(B)** Distributions of the mean correlations in (A) across samples, shown for selected regulons for five MPs (each shown in a separate panel).

Major point 4

“One major comment I want to make here is: the manuscript needs more necessary validations (i.e., RNAScope and IHC staining) on the findings, such as MPs and associations between TME cell types in various cancer types. I do appreciate the NMF method has been proven robust in previous studies, but some experimentally validations are still necessary to corroborate findings such as (but not just) MP30 in non-PDAC cancers.”

We agree with the reviewers that additional validation using methods orthogonal to those used in the original manuscript is important for enhancing the credibility of our results. We note that the primary objective and focus of this work is to provide a *broad* description of the expression patterns of intra-tumor heterogeneity, which can only be done computationally based on an extensive compendium. However, we agree that for some specific results, such as the presence of MP30 in non-PDAC tumors, it is important to also provide an experimental or orthogonal support. To this end, we focused on recapitulating findings from the scRNA-seq data using three modalities:

- (1) **IHC staining:** we performed extensive IHC staining with various antibodies and in six cancer types and validated a range of observations from the scRNA-seq analysis, including MP30 in non-PDAC cancers (**Response Figure 6-9**).
- (2) **Spatial proteomics:** we downloaded CODEX datasets from three independent recent studies of HNSCC and CRC tumors. Using this approach, we were able to validate the proposed synchronization in cell cycle between adjacent cells of distinct cell types (**Response Figure 10**).
- (3) **Spatial transcriptomics:** We downloaded VISIUM datasets from three independent recent publications of ovarian, skin and glioma tumors. Using this approach, we were able to validate the existence, abundance and interaction across several MPs (**Response Figure 11**).

Notably, approaches 1-2 are orthogonal to scRNA-seq, while Visium data is composed of spatial RNA-seq data generated in a different platform compared to the standard scRNA-seq methods and is generally not at single-cell resolution. Complementary to the methods above, some of our findings were also consistent with analysis of TCGA bulk data, such as detection of PDAC-classical MP in non-PDAC tumors.

IHC staining

As an orthogonal validation, we performed IHC staining of FFPE sections across a variety of tumor types (breast ductal adenocarcinoma; lung squamous, lung adenocarcinoma and invasive mucinous lung cancer; head and neck squamous cell carcinoma – laryngeal and oropharyngeal; pancreatic ductal adenocarcinoma), each with multiple blocks.

First, we validated that the EpiSen and PDAC classical MPs, each of which were initially described in a single tumor type, are indeed observed across multiple cancer types. The EpiSen MP was originally described in HNSCC tumors and cell lines and validated by SPRR1B staining (Puram et al. 2017, Kinker et al. 2020). We used the same SPRR1B antibody and indeed found subsets of positive cancer cells also in squamous lung carcinoma, lung adenocarcinoma, and breast cancer (**Response Figure 6A**). We identify two flavors of EpiSen: one which also includes a

senescence-associated secretory phenotype (SASP)-like component (SPRR1B+SLPI+) and one without the secreted component (SPRR1B+SLPI-)(**Response Figure 6B**).

Similarly, the PDAC classical program, originally associated with pancreatic cancer, is also identified in invasive mucinous lung adenocarcinoma, depicted by a subset of panCK+ cells that highly express the previously used PDAC classical marker TFF1 which is the #1 gene in MP30 (**Response Figure 6C-D**). In both PDAC and invasive mucinous lung adenocarcinoma we see that the subset of panCK+ cells co-expressing TFF1 is spatially separated from other cancer cells.

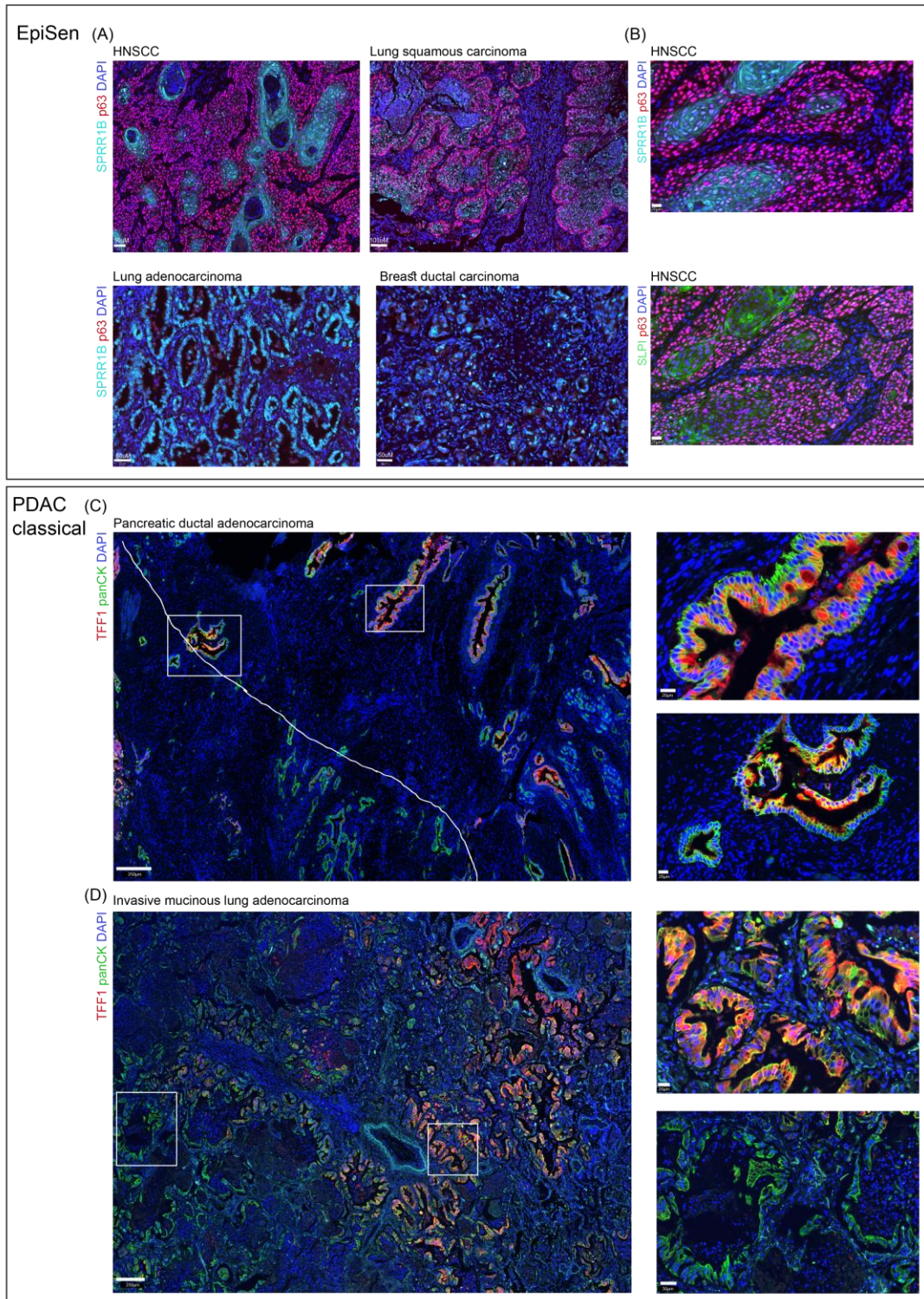
Second, we demonstrate that EpiSen represents an aberrant form of normal epithelial differentiation in HNSCC, consistent with our results in main **Fig. 4C**. Normal differentiating epithelium forms organized structures of basal layer p63+ cells followed by a more differentiated and less proliferative SPRR1B+ layer (**Response Figure 7**). Juxtaposed within the same image field we find malignant structures in the tumor mimicking normal epithelial differentiation with a layer of SPRR1B+ p63- cells atop vast numbers of p63+ cells but with increased disorganization compared to the normal structures (**Response Figure 7**).

Third, IHC was used to validate aspects of the Interferon-response/MHC-II MP. In new analysis that we have added to the revised manuscript we examine the coupling between MHC-II and interferon response, demonstrating that coupling is largely specific to cancer cells, and even among cancer cells we can find both coupled and decoupled programs. To validate these observations we performed IHC with a combination of an MHC-II marker (HLA-DRA), an interferon response marker (ISG20) and cell type markers.

Unlike in non-malignant cells, where interferon and MHC-II expression are typically decoupled (e.g. **Response Figure 8A** left upper panel, asterisk), we find subpopulations of malignant cells that co-express MHC-II and interferon response (MHC-II+ ISG20+) in squamous lung carcinoma, breast cancer, lung adenocarcinoma, and HNSCC (**Response Figure 8**). We observed that interferon response and MHC-II in cancer cells often form a spatial expression gradient, with subsets of malignant cells expressing both markers to varying degrees, and with highest expression for one marker vs. the other on the extreme ends of the gradient (e.g. **Response Figure 8C, right panel**). Additionally, we frequently observed clusters of T cells in close proximity to cancer cells expressing interferon response (**Response Figure 8D**).

Finally, we used IHC to validate an interesting interaction from the scRNA-seq analysis, between the complement MP in fibroblasts and the high endothelial venules (HEV) MP in endothelial cells (**Fig. 5A**). As expected, we observe perivascular fibroblasts (SMA+) expressing complement (C3+) around MECA79+ HEVs (**Response Figure 9**).

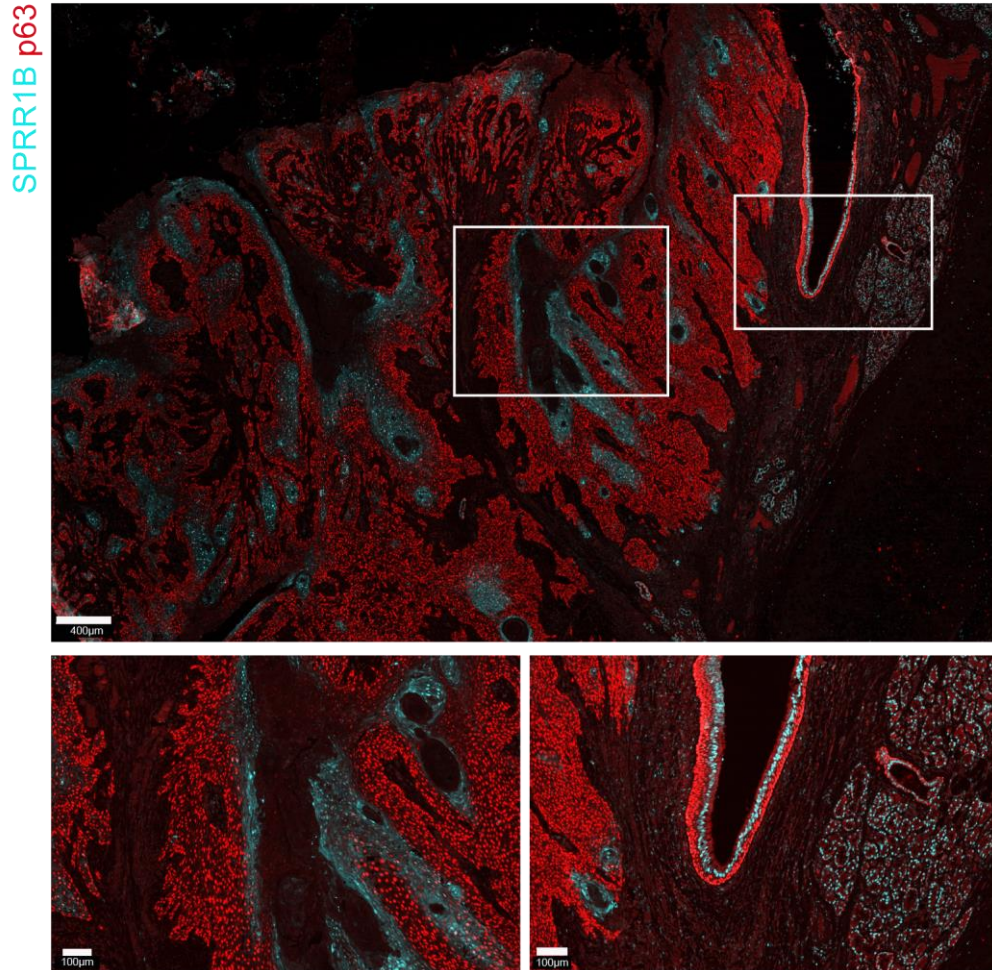
IHC results are described in the manuscript in **pages 8, 11 and 13**, and shown in **Fig. 3C,E, Fig. 4F, Fig. 5D and Fig. S11-12, S21 and S24**.



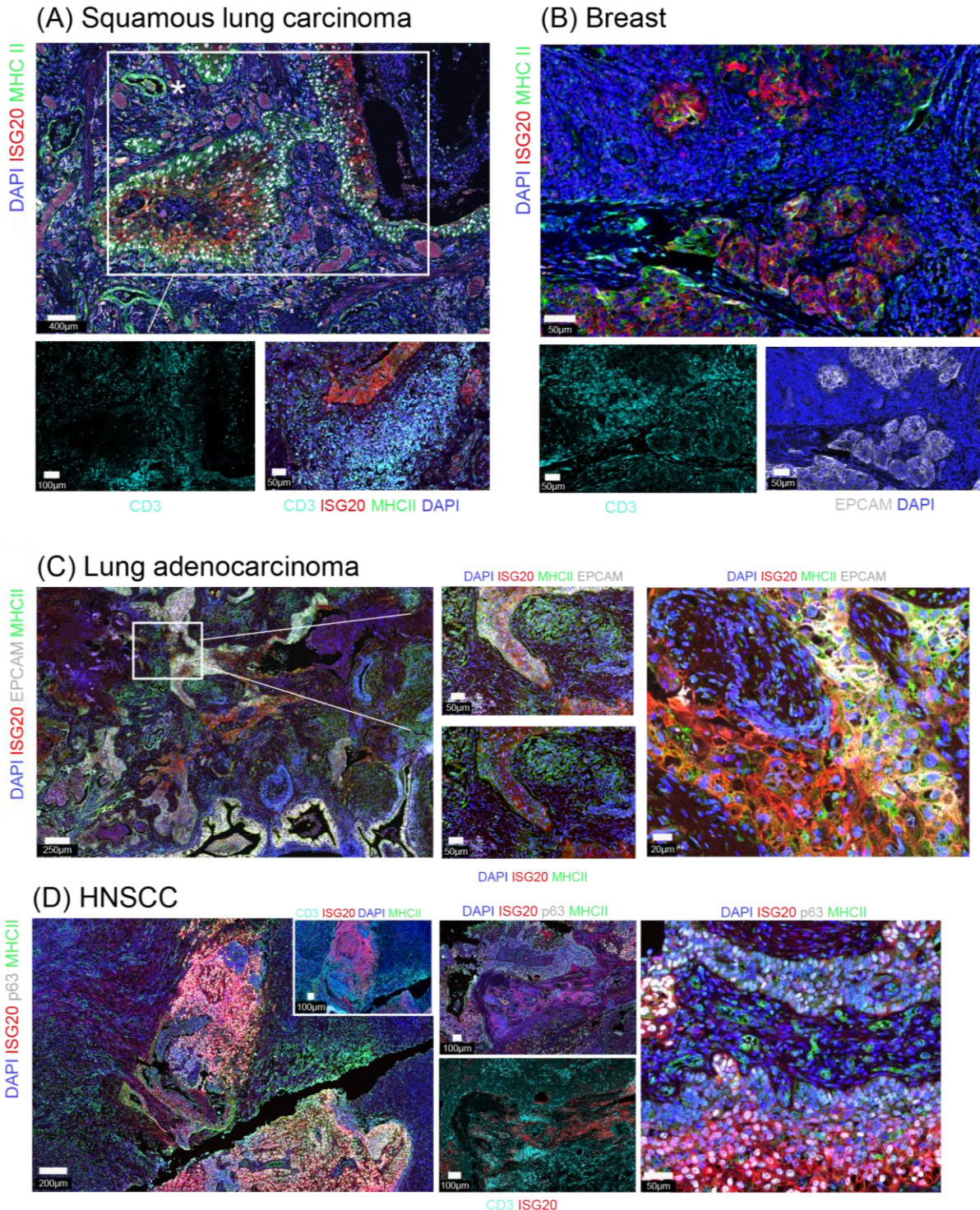
Response Figure 6. Experimental validation related to EpiSen and PDAC-classical MPs. (A) Example of EpiSen in HNSCC (SPRR1B+) with the senescence-associated secretory-like phenotype (SLPI+). **(B)** Example of the EpiSen flavor with the senescence-associated secretory phenotype expressing both SPRR1B and SLPI. **(C-D)** The PDAC-classical MP (panCK+TFF1+) is also observed in other cancer types such as invasive mucinous lung adenocarcinoma. In both PDAC and invasive mucinous lung adenocarcinoma subsets of cancer cells expressing TFF1 are spatially zoned.

EpiSen

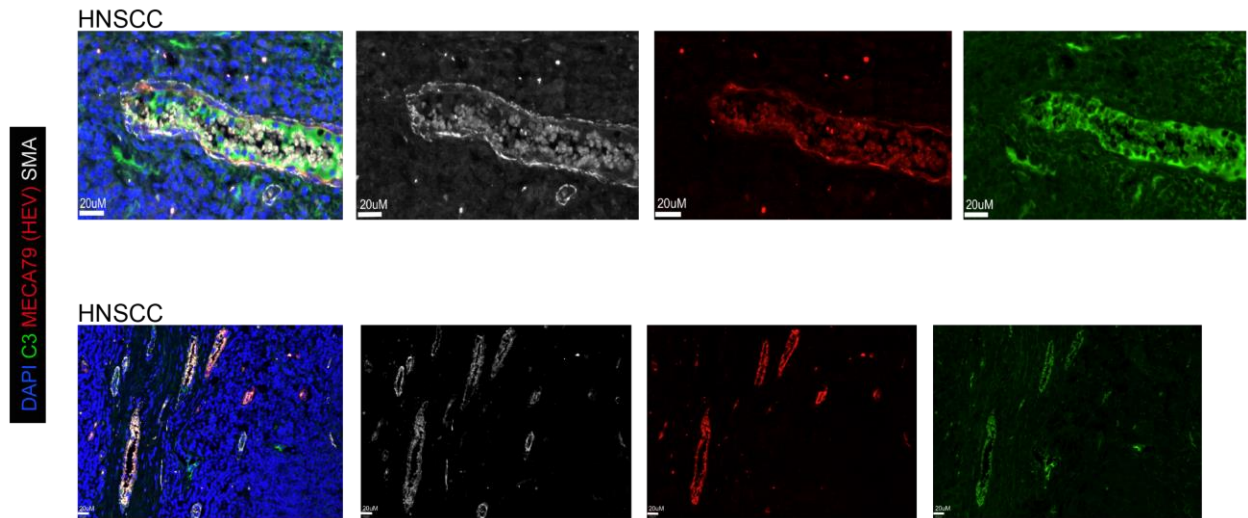
HNSCC



Response Figure 7. EpiSen represents an aberrant form of normal epithelial differentiation. Right lower panel shows layers of normal differentiating epithelium with an organized layer of SPRR1B+ cells adjacent to p63+ cells. Left lower panel reveals a disorganized malignant version of a similar structure.



Response Figure 8. Coupling and decoupling of MHC-II and Interferon response. In malignant cells, MHCII and interferon response are often co-expressed, unlike in non-malignant cells where they are uncoupled (e.g. asterisks in A, right panel in D). Cancer cells co-expressing MHC-II and interferon response frequently form an expression gradient in which the cancer cells co-express both markers on a continuum (upper panels in A and B; panel C; left panel in D). T cells often cluster around cancer cells with high expression of interferon response and MHC class II (left lower panels in A; left lower panel in B; left and middle lower panels in D).



Response Figure 9. Experimental validation of interaction between the fibroblast complement MP and the endothelial HEV MP. High endothelial venules (MECA79+) are surrounded by perivascular fibroblasts (SMA+) with upregulated expression of complement (C3+).

CODEX

Antibody based highly multiplex immunofluorescence imaging (i.e. CODEX) is an ideal orthogonal technology to that used in our study because it detects proteins (as opposed to RNA) and has true single cell resolution with spatial information. This makes it possible to stain conventional fresh frozen or FFPE tissue sections, annotate cell types and investigate functional markers (i.e. Ki67 for proliferation) with a spatial dimension. We therefore downloaded several CODEX datasets and reanalyzed them.

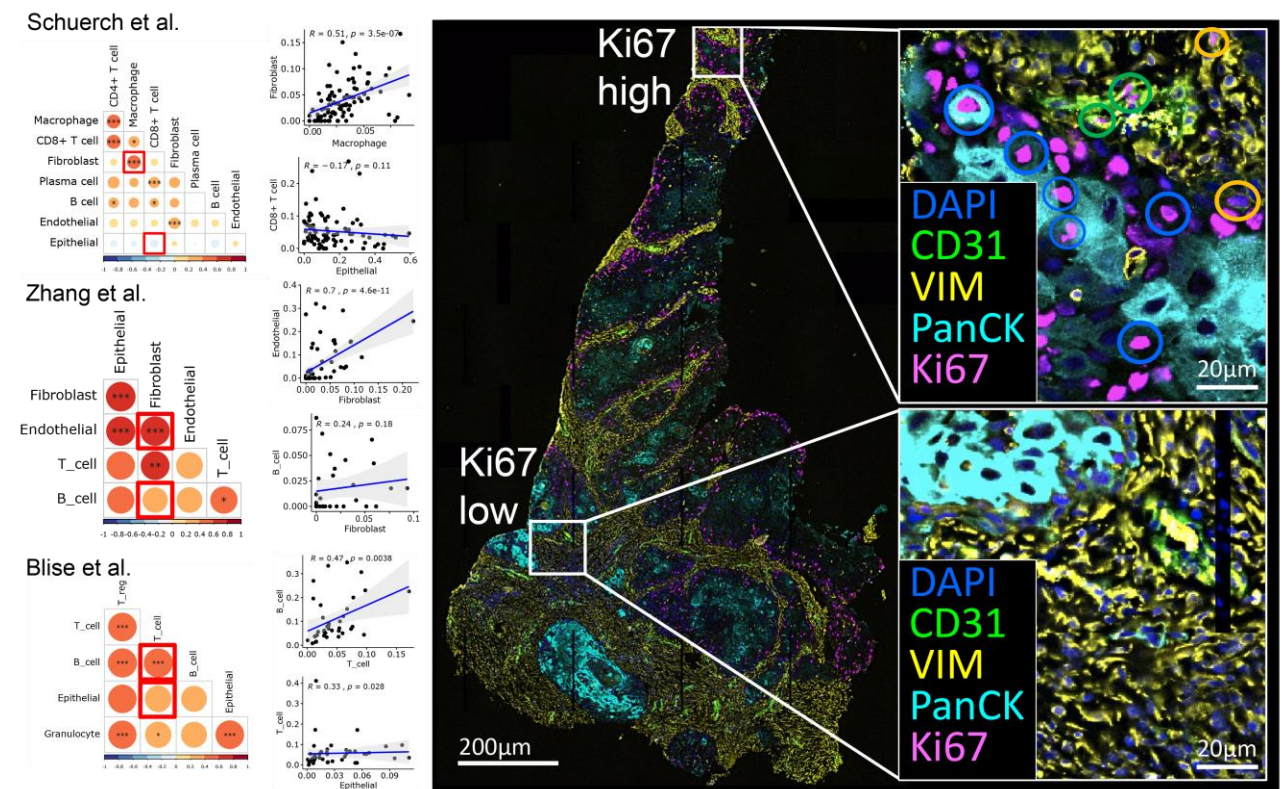
The main limitation is that existing CODEX datasets use a limited set of markers that do not adequately address most questions that we would like to examine. We therefore focused our attention on proliferation. As depicted in **Fig. 5A**, one major finding was the correlation among cell-cycle MPs of different cell types, suggesting non-intrinsic effects driving proliferation of both malignant and non-malignant cell types. To validate this result, we took advantage of three public available CODEX datasets on epithelial cancers. These comprised two studies on HNSCC (Blise et al. Nature precision oncology 2022, Zhang et al. Nature methods 2022) and one on CRC (Schuerch et al. Cell 2020) resulting in a total of 243 annotated regions of interest coming from 59 tumors and 50 patients. We assessed Ki67-indices reflecting the proportion of cycling cells for every annotated cell type and performed correlation analysis within all regions of interest (n=243, area=2300-25002µm²).

As depicted in **Response Figure 10** we found significant positive correlation in all three studies between cycling B cells and T cells and between cycling fibroblasts and endothelial cells in the two studies that covered these cell types (Zhang et al., Schuerch et al.). This is in line with main Figure 5A that shows the strongest correlation of the cycling MP in these two pairs of cell types.

Moreover, in the CODEX data-sets we identify additional positive correlations between macrophages and T cells, T cells and plasma cells (Schuerch et al.), epithelial (as approximation for malignant) cells and fibroblasts, T cells and fibroblasts (Zhang et al.), epithelial (as approximation for malignant) cells and granulocytes (Blise et al.). **Response Figure 10** shows the corresponding correlation plots, together with representative examples (scatter plots) for each study, showing positive correlations and no correlations for pairs of interest, respectively.

The immunofluorescence image on the right depicts a whole HNSCC section. Areas of high proliferation (Ki67 high) vs. low proliferation (Ki67 low) are highlighted. The upper magnification boxes show a general proliferative state of malignant (epithelial) and non-malignant cells (endothelial, fibroblast) in close spatial proximity. In contrast, the magnification boxes on the bottom magnify a Ki67 low area, where none of these cell types are cycling.

Taken together, we validated that epithelial tumors show intra- and inter-tumor heterogeneity related to areas of high vs. low proliferation. Surprisingly, the proliferating cell types in Ki67 high areas comprise not only cancer but also cells from the TME, suggesting the presence of high proliferative malignant and non-malignant niches that are driven by secreted or other external factors.



Response Figure 10. Experimental validation using CODEX data. Left panels show quantification of the correlation between cycling cells of different types across studies, together with representative examples (scatter plots). Right panel is an immunofluorescence image of a HNSCC section, highlighting a Ki67-low region (bottom) and a Ki67-high region (top). Right panels: Top box depicts cycling cells (Ki67-high niche) of different cell types that are in spatial proximity, compared to non-cycling cells in the bottom box (Ki67-low niche). PanCK (blue), CD31 (green) and VIM (yellow) mark malignant, endothelial and fibroblast cells, respectively.

The CODEX analysis is described in the manuscript in **page 12** and shown in **Fig. 5B** and **Fig. S23**.

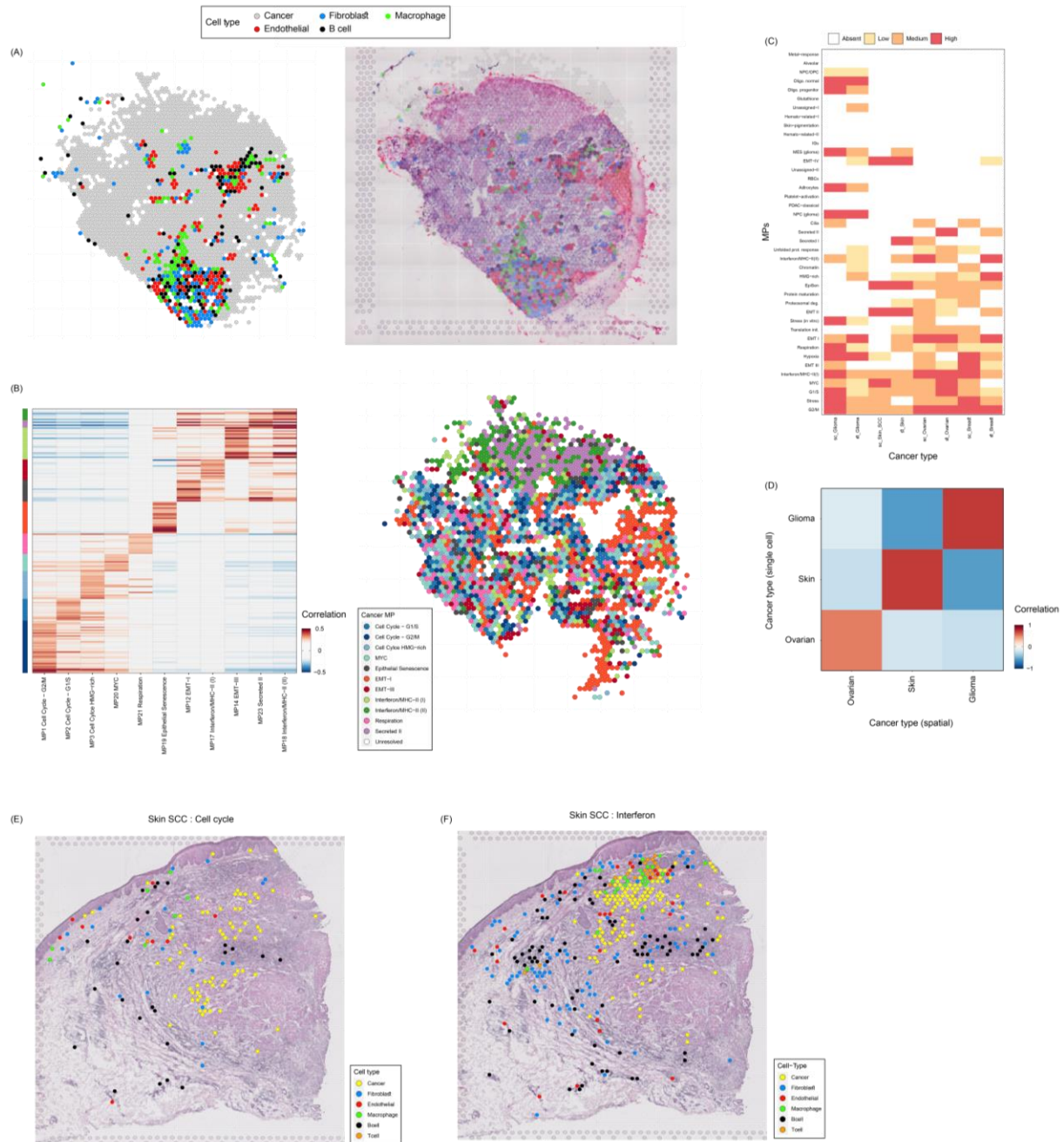
VISIUM

The 10x Genomics Visium platform is an alternative approach to analyze intra-tumor heterogeneity that differs from scRNA-seq in several important ways, including lack of tissue dissociation and retaining spatial information. Unlike CODEX, which is a targeted approach requiring a priori selection of antibodies, Visium enables profiling of the whole transcriptome, although at a more limited resolution, with few cells per spot. We used previously published, publicly available Visium datasets, spatially profiling tumor samples from ovarian cancer (8 samples), skin squamous cell carcinoma (4 samples), and glioblastoma (9 samples). For each Visium sample, we first inferred CNAs to assign spots as malignant or non-malignant, and then assigned the non-malignant spots to cell types based on expression of canonical cell type marker genes (**Response Figure 11A**). Although each spot likely contains multiple cells, this analysis is aimed at uncovering the main cell type in each spot.

Next, spots that were confidently assigned as malignant were scored for the scRNA-seq-derived cancer MPs. Similarly, spots of non-malignant cell types were scored for the MP of the respective cell type. An MP was defined as detected in a given sample if at least 10 of the MP genes had a Pearson correlation above 0.2 with the MP scores (**Response Figure 11B**). Notably, in each of the three cancer types, we found largely the same MPs by scRNA-seq and by Visium, supporting our scRNA-seq identification of MPs and their cancer-type specificity (**Response Figure 11C-D**).

We further utilized the Visium data to examine the spatial proximity between cellular states that were identified as interacting in the scRNAseq analysis. We did identify spatial patterns that support some of the proposed interactions, such as proximity between cell cycle states and interferon-response states of distinct cell types (e.g. **Response Figure 11E-F**). However, due to the non-single-cell resolution of this platform, we felt that these results remain difficult to interpret and therefore we do not focus further on these results.

The identification of cancer MPs in Visium data, and the similar distribution of MPs across three cancer types, between scRNA-seq and Visium, are described in the manuscript in **page 8** and shown in **Figure S9**.



Response Figure 11. Analysis of MP expression in spatial transcriptomics data. (A) Visium image of an ovarian sample showing the spatial organization of spots assigned to different cell types. **(B)** Correlation between overall MP scores and specific MP genes (rows), for MPs (columns) with at least 10 genes with correlation above 0.3. Correlations were correlated across spots of the ovarian cancer sample shown in A; right panel shows the same sample, with assignment of malignant spot to MPs based on maximal score. **(C)** abundance of MPs in scRNAseq data (denoted as 'sc') and in spatial transcriptomics data (denoted as 'st'), for glioma, skin and ovarian cancer. Abundance categories are defined as in **Fig. 3A**. **(D)** Correlations between the single-cell and the spatial-transcriptomics columns in (C), demonstrating consistency in the identification of MPs in each of the three cancer types. **(E-F)** Co-localization of spots assigned to the cell cycle (left) and Interferon response (right) MPs, for malignant and TME spots in a selected skin sample.

Major point 5

“The authors conducted comparison between MPs in malignant cells and those in non-malignant epithelial cells. It is interesting to see that most MPs showed high similarities between the two cell types, which implies these common MPs are operative throughout the process of malignant transformation. However, for those malignant-specific or oncogenic MPs, it would be interesting to further look into the time points of their occurrence during tumorigenesis. Is there any data from pre-cancerous tissues that can be potentially integrated and analyzed? It could be interesting to compare MPs in normal epithelial cells, pre-cancerous epithelial cells and cancer cells. Similarly, the comparison between primary and metastatic tumors can also be of interest. Is there any data from metastatic tumors (lymph node or distant metastasis) can be integrated and analyzed? Or can the authors check MPs in bulk RNA-seq data from pre-cancerous tissues and metastatic tumors?”

We divide our response into the following sections on which we elaborate below:

- (1) Non-malignant tissue - We curated normal scRNA-seq data from 184 samples originating from 24 organs that included ~740,000 normal cells in total. We generated MPs for the main normal microenvironment cell types and compared them to their malignant counterparts. We found that many MPs from tumors (in malignant and in TME cells) are recapitulated also in non-malignant tissue, but some programs are more pronounced in tumor tissue (i.e. are seen exclusively or more abundantly in cancer samples compared to non-cancer), and vice versa.
- (2) Pre-cancerous tissue – Published data of this kind was limited. Focusing on colon data, we inferred MP expression in metaplastic (adenoma) data and compared their distribution to that in malignant and normal colon tissue. We found that adenoma tissue resembled malignant tissue (compared to normal tissue) in their MP expression in a few MPs, whereas there were MPs in which the opposite was true.
- (3) Metastatic tissue - Sufficient data for a comprehensive comparison using scRNA-seq was lacking. We considered the possibility of analyzing bulk RNA-seq data but concluded that the confounding effects of primary vs. metastatic tumor compositions would make it difficult to obtain any robust results.

Non-malignant tissue

Similar to the way we compared MPs derived from non-malignant cells within malignant tissues to MPs derived from malignant cells (main Figure 4), here we expanded the analysis by performing the comparison to MPs derived from normal tissue. Our goal was to determine which MPs are unique to the malignant or normal-tissue context, and which are expressed and distribute similarly.

To this end, we curated published normal scRNA-seq data from 4 large studies that together included 184 samples and 741,309 cells. We then followed a similar approach to how we defined MPs in malignant tissues, in which we verified the published cell annotations, derived robust NMF programs and finally generated MPs for epithelial cells, T-cells, B-cells, macrophages, fibroblasts and endothelial cells (see Methods for more details). In the last step of generating MPs, for each cell type we integrated the robust NMF programs from tumor and normal tissues before the

clustering was done, in order to test which clusters were unique enriched in one tissue type, and which clusters were comparable between tumor and normal tissues. As an alternative, we also derived MPs only with the NMF programs from normal tissues and these are provided in **Table S11**.

Response Figure 12A-F summarize the results for the six microenvironment cell types. Each panel consists of a main heatmap of clustered NMF programs, with the following bars above it (from top to bottom):

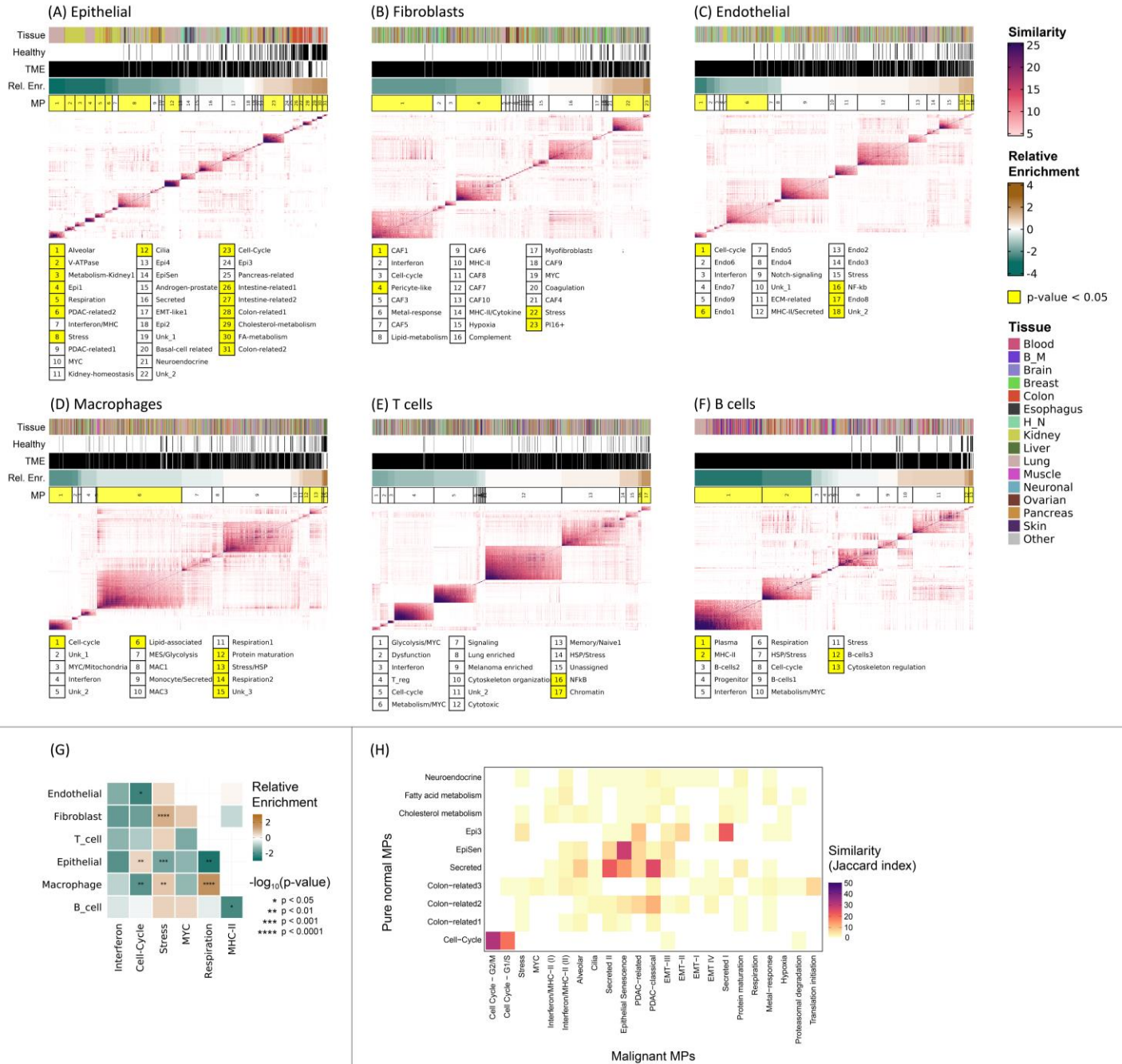
- (i) Tissue of origin.
 - (ii) Healthy vs. cancerous tissue (two separate bars).
 - (iii) Relative enrichment score - log ratio between observed and expected number of normal NMF programs per cluster.
 - (iv) Cluster borders, with yellow indicating a statistically significant relative enrichment score.
- Annotations were given to each cluster/MP based on functional enrichments, whereas MPs that were suspected to represent cells of low quality or miss-annotations were removed.

Overall, we find that nearly 70% of the MPs are comparable between cancer and normal tissues, which we define as MPs that were not statistically enriched for cancer or normal NMF programs. The fraction of comparable MPs is particularly high in T-cells (88% of MPs) and low in epithelial cells (51% of MPs).

Response Figure 12G shows the relative enrichment of several shared MPs in normal relative to tumor tissues. Cell cycle was depleted in most cell types, in normal tissues relative to their tumor counterparts, consistent with our interaction analysis in **Fig. 5A** (which suggested that cell cycle is correlated across non-malignant cell types within the tumor and hence is potentially upregulated in tumors). Epithelial non-malignant cells were the exception as they showed the opposite pattern. Further inspecting the origin of cycling epithelial cells revealed that they were enriched in colon and small intestine tissues, which harbor a high degree of cycling cells under physiological conditions. The stress MPs were overall enriched with normal cells, except for epithelial cells, perhaps indicating their susceptibility to tissue processing.

Finally, as noted above, we also derived MPs from the normal NMFs without prior integration with the malignant NMFs. This enabled us to directly compare the MPs of epithelial cells in normal tissues (i.e. pure normal MPs) to the malignant MPs (**Response Figure 12H**). In this analysis, we find more limited similarity - compared to the analysis of non-malignant epithelial cells within tumors (**Fig. 4C**) - suggesting that some of the malignant MPs are activated also in non-malignant epithelial cells but only within the tumor context.

The similarities and differences of MPs between normal and tumor tissues is noted in the manuscript in **page 10** and shown in **Fig. S17** and **Table S11**.



Response Figure 12. Meta-programs in normal (healthy) vs. tumor tissues. (A-F) Each panel shows the clustering of NMF programs to define MPs in one cell type. NMF programs were derived from both normal and malignant datasets. Heatmaps show their similarities (by Jaccard index) and the clustering that defines MPs (with MP borders shown in the panel above). Top panel also shows the origin of each NMF program – the tissue type and whether it is from healthy or tumor (TME) sample, and the relative enrichment of each cluster (i.e. MP) in healthy vs. tumor samples. Enrichment was calculated as $\log_2(\text{observed \# NMFs} / \text{expected \# NMFs})$ and corresponding P-values were calculated by hyper-geometric test and corrected for multiple comparisons using FDR. Significant enrichments are highlighted in yellow. MPs and their annotations are listed below each heatmap. **(G)** Relative enrichments in normal vs. tumor tissues of the MPs that are observed across different cell types. **(H)** Overlap between selected malignant MPs and MPs derived using solely normal NMFs.

Pre-cancerous tissue

We next sought to determine to what degree MP expression varies along the process of tumorigenesis. To this end, we focused on colon/intestine tissue in which pre-malignant stages are well defined, and for which there was sufficient published data. We found that, overall, pre-malignant adenoma tissues express intermediate MP levels, between those of normal and malignant tissues.

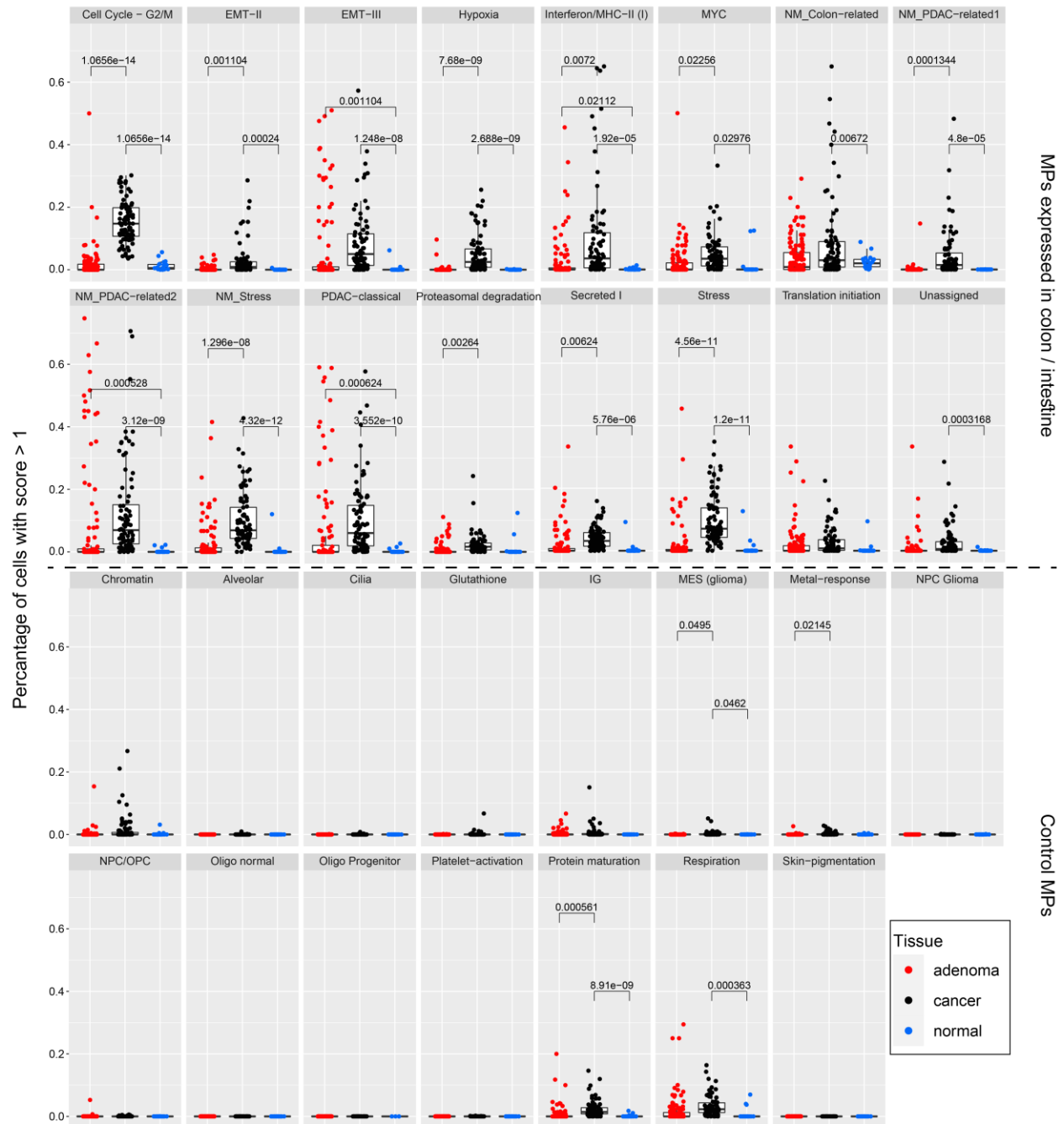
Specifically, we integrated the following scRNA-seq data:

- (i) Non-cancerous tissue obtained from the Burclaff et al. 2022 and the Jones et al. 2022 studies that were also used in the analysis above of normal tissues.
- (ii) Pre-malignant adenoma tissue (serrated and adenomatous polyps from Chen et al. 2021) that included 19,312 cells from 76 samples with at least 20 cells per sample.
- (iii) Malignant colon tissue from two studies that were used in the main text for deriving the malignant MPs (Lee et al. 2020 and Li et al. 2017) along with a recent large new published dataset (Pelka et al. 2021) that included 108,999 malignant cells from 64 colon samples and will be added to the next version of 3CA.

We first selected the MPs that were expressed in malignant and non-malignant epithelial colon cells according to main **Fig. 3A** and **Fig. 4B**. The malignant MPs included Cell Cycle - G2/M, Stress, MYC, Interferon/MHC-II (I), EMT-II, EMT-III, Hypoxia, Translation initiation, Proteasomal degradation, Secreted I, PDAC-classical and the Unassigned MP. The MPs obtained from non-malignant epithelial cells within tumors included: Stress, PDAC-related1, PDAC-related2 and Colon-related (**Table S10**).

After standard quality control (described further in the Methods section), we centered the epithelial per study, scored each sample to the MPs listed above, and calculated the percentage of cells (within each sample and per MP) with score larger than 1. The distributions of these percentages are shown in **Response figure 13** (top boxplots). The cancer samples were significantly higher than the pre-cancerous and normal samples in most of the cases, whereas pre-cancerous and normal samples were overall similar except for the Interferon, EMT-III and PDAC-classical MPs in which the adenoma samples were higher. To test whether the results above might have been influenced by technical differences between the distinct types of samples, we performed a similar analysis using 'control' MPs (lower boxplots) which were not identified in colorectal cancer samples. As expected, there were much less differences between the distributions of the different samples, supporting the relevance of the results in the upper plots.

The pre-malignant analysis is mentioned in the manuscript in **page 10** and shown in **Fig. S18**.



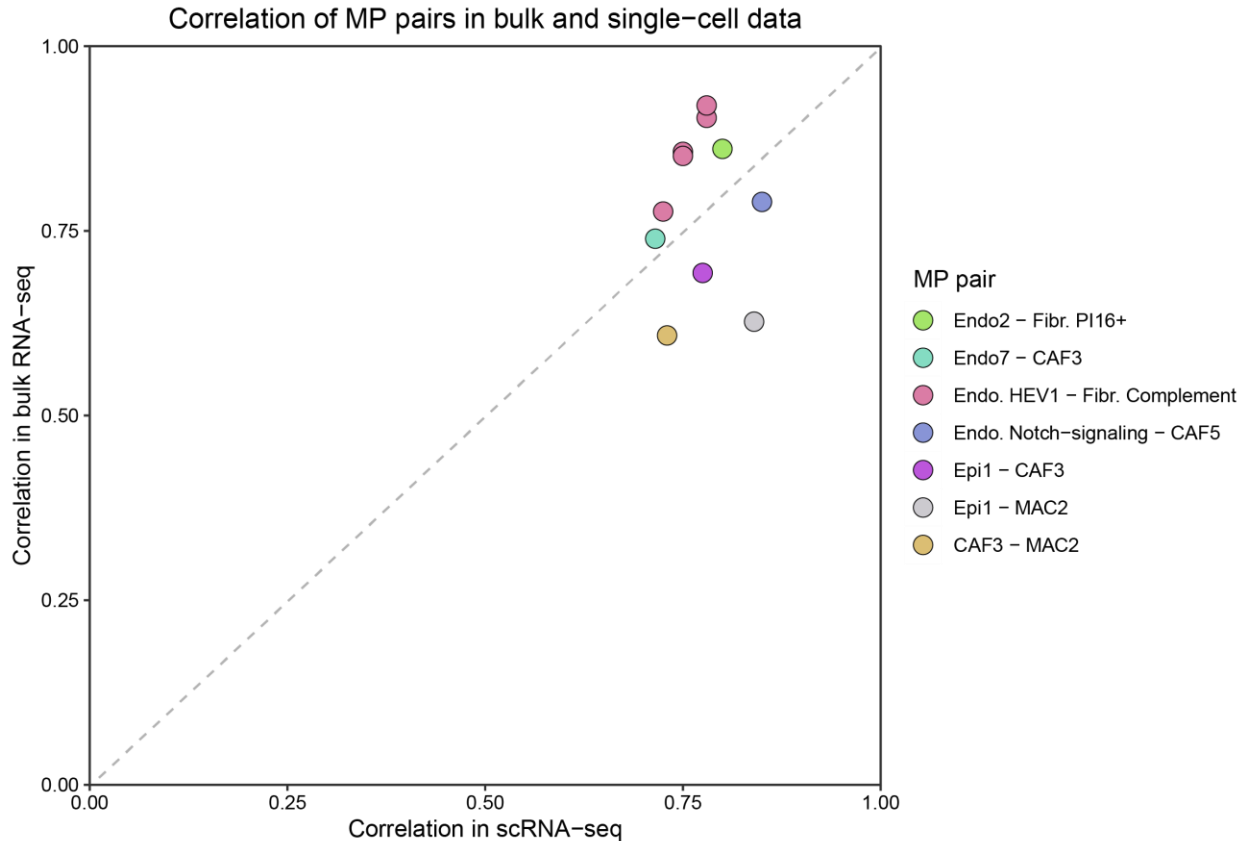
Response Figure 13. MP expression in pre-cancerous tissue. For each MP (panel), box plots show the percentages of cells that scored above 1 in pre-malignant (red), malignant (black) and normal (blue) tissue samples. MPs above the dashed line are those detected in colon and those below the lines were not detected in colon and are added as a control for technical differences between tissue types. P-values were calculated by t-test with bonferroni correction. Only significant cases are highlighted by showing the p-value. 'NM' pasted at the beginning of some MP names stands for non-malignant.

Major point 6

“The authors inferred potential associations between cell types in TME by examining the co-occurrence between MPs of different cell types. Can the authors further validate this in larger cohorts by using bulk RNA-seq data (i.e., through deconvolution)?”

Validation of these observations using bulk RNA-seq data is highly challenging due to (i) the high degree of similarity between related MPs from different cell types (e.g. overlapping interferon responses and cell cycle programs are seen across all cell types), and (ii) similarities between MPs and cell type signatures (e.g. EMT programs and fibroblast signature). Both of these issues can easily confound associations between MPs. Deconvolution methods attempt to resolve this problem, but they are still limited to approximate proportions and expression profiles for relatively few coarse cell types. While we did obtain a matrix of inferred cell-type-specific expression profiles obtained by the deconvolution algorithm CIBERSORTx, its expression profiles for rare cell types are limited to a relatively small number of genes, making it unsuitable to quantify MPs from TME cell types. The limitations of such deconvolution algorithms were a key motivation for a recent paper from our lab (<https://doi.org/10.1038/s41467-021-22800-1>), in which we went to great lengths to decouple EMT from stromal confounders in bulk RNA-seq data, an approach that cannot practically be applied to all MPs defined here.

Given these technical problems, we attempted to partially address the reviewer’s question by restricting the analysis to those MP pairs that are least likely to be confounded by other MPs or cell types. For this, we isolated pairs of MPs for which we observed significant associations in the scRNA-seq cohort but which have small pairwise overlap, and in which each MP has small overlap with all other MPs (see **Methods**). For these few interactions, the correlations in scRNA-seq data agreed well with those in bulk RNA-seq data (**Response Figure 14**). However, this analysis remains limited and could not be extrapolated to all MPs. Overall, we argue that bulk RNA-seq is not wholly appropriate for validating our observed MP interactions, and that this task may be better approached in the near future once we have expanded 3CA with more scRNA-seq datasets, as well as spatial transcriptomics data.



Response Figure 14. Correlation between MPs in bulk and single-cell RNA-seq data. Scatterplot with each point corresponding to a pair of MPs (indicated by color) observed in one cancer type, such that multiple points of the same color reflect the same MP pair observed in different cancer types. MP pairs were selected for having significant correlations in scRNA-seq data while having small pairwise overlap and small overlap with all other MPs. For each point, representing a combination of MP pair and cancer type, the x axis value measures the average Pearson correlation of these MPs across the relevant scRNA-seq datasets for that cancer type, while the y axis value measures the correlation of these MPs in TCGA bulk RNA-seq datasets for that cancer type.

Major point 7

“In the current manuscript, common MPs (i.e., EMT programs) were found to be shared by multiple cancer types (Figure 3A) and cell types (Figure 4C). In addition to discussing the similarities, can the authors also have more discussion regarding differences of same MPs in different cancer types and cell types?”

To address this point, we searched for differentiating features between the same MPs in different cancer types and related MPs in different cell types.

Same MPs in Different cancer types

We systematically investigated the unique profiles of each MP in each cancer type in which it was detected in at least 10 samples. In each case, we compared the profiles of a given MP in a given cancer type to the same MP across all other cancer types (as long as there were at least 10 samples in the other cancer types). We used two complementary methods for comparisons:

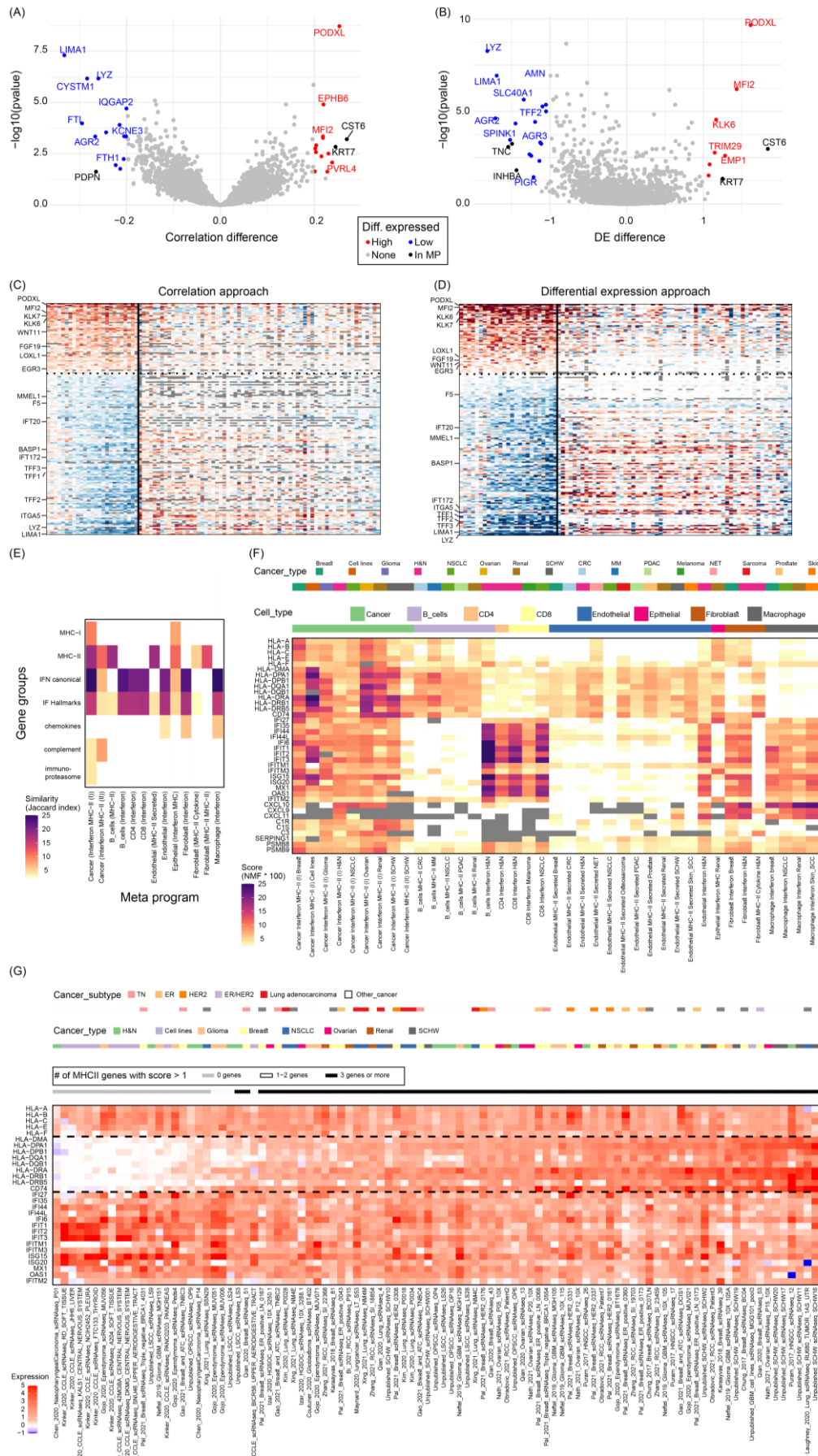
- (1) Correlation-based: We calculated the correlation of each gene with the scores of the MP within each sample, then averaged the correlations per cancer type and compared these average correlations.
- (2) DE-based: We calculated the differential expression between cells with MP scores above 1 and cells with MP scores below 0 within each sample, then averaged the differential expression per cancer type and compared the average DE values.

For example, results from the two methods are shown for the EMT-II cancer MP in PDAC vs. non-PDAC samples (**Response Figure 15A-D**). The two methods had high overall agreement and in subsequent analysis we included only genes that were upregulated or downregulated in both (**Table S12**).

Related MPs in different cell types

The overall similarities (i.e. overlapping genes) in comparisons of related MPs from different cell types are shown in **Fig. 4C-D**, **Fig. S16** and **Fig. S19**. In addition, we highlight in the manuscript (pages 10-11) several specific cases with notable differences related to coupling/decoupling within MPs:

- (1) The EpiSen malignant MP harbors both epithelial differentiation genes and secreted factors, while these are observed as two distinct MPs of non-malignant epithelial cells (**Fig. S19B**). This suggests that such differentiation and secretion are only coupled in the malignant cells, possibly reflecting an aberrant senescence response.
- (2) The Alveolar MP of non-malignant cells includes multiple MHC-II genes, consistent with the known antigen presentation by normal lung AT2 cells. Yet, the highly similar malignant Alveolar MP lacks MHC-II genes, suggesting that specifically in malignant cells, alveolar differentiation is decoupled from antigen presentation (**Fig. 4D**, **Fig. S19A**).
- (3) In malignant cells interferon response and MHC-II were tightly coupled within the same MP (called interferon/MHC-II), while we did not observe such coupling in the TME cells. Aside from the epithelial cells, we observed only distinct MHC-II MPs and/or Interferon MPs in all TME components (**Response Figure 15E-F**). Moreover, among the cancer cells we found differences in the degree of coupling (between interferon and MHC-II) across samples and cancer types (**Response Figure 15G**). These results were further supported by IHC analysis that demonstrated coupling in malignant cells of multiple cancer types, but not in non-malignant cells, and in some cases a gradient from coupled to decoupled malignant cells (**Response Figure 8**).



Response Figure 15. Dissimilarities within MPs across different cancer types and cell types. (A) Comparing the EMT-II MP between PDAC samples and samples from other cancer types. For each gene, the correlation with the MP scores were calculated in each sample with the MP, and those correlations are compared between PDAC and non-PDAC samples, showing their mean (X-axis) and significance (Y-axis). Red and blue reflect genes with significantly higher or lower MP correlations in PDAC, respectively. **(B)** Same as A, but instead of correlation with MP, the measure for each gene in each sample is the differential expression (log-ratio) between cells with MP score above 1 and those with score below 1. **(C,D)** heatmaps corresponding to A and B that show the actual correlation or differential-expression values across the PDAC samples (to the left of the black vertical line) vs. the other samples, showing genes that were high in both approaches (above the dashed line) vs. genes that were low in both approaches. The genes in both heatmaps are the same but are sorted differently (in decreasing order according to their average in PDAC samples). **(E)** The overlap between MHC-II or Interferon related MPs from different cell types and several curated gene-sets. The only MP with considerable overlap to both MHC-II and Interferon response came from cancer. **(F)** NMF scores of samples that participated in E averaged according to cancer type. Grey indicates missing values. **(G)** Average expression of MHC and interferon response genes (rows) across samples (column) in which the Interferon/MHC-II(I) MP was detected. In each sample, the expression was averaged only among cells that scored above 1 to the MP. Samples are sorted by the average difference between MHC-II and Interferon genes. Genes above the top broken line reflect MHC-I, genes between the broken lines reflect MHC-II, and genes below the bottom broken line reflect Interferon response.

Minor point 1

“Can the authors further discuss global cell state and relative cell state, and how they influence tumor behaviors?”

We added the following text to the discussion:

“The transcriptome of a cancer cell may be considered as a proxy to its “global cell state”. Comparison of global cell states across patients underscores a high degree of cellular heterogeneity in cancer that remains difficult to interpret. We argue that dividing the differences in global cell states to those that reflect inter-tumor and those that reflect intra-tumor heterogeneity (i.e. the “relative” cell state within a tumor) helps to make sense of such high heterogeneity. Inter-tumor differences reflect the cumulative effects of genetic and epigenetic aberrations that have been acquired during the lengthy oncogenic process of each tumor, thereby creating distinctive tumor-specific profiles; while intra-tumor differences primarily reflect more recent events that have shaped the state of cells, including their phase along the cell cycle, their short-term responses to cytokines and nutrients (or lack thereof), and their instantaneous cellular state with regards to stochastic fluctuations in gene expression. While inter-tumor differences have been widely studied during the last two decades through bulk RNA-seq profiles, for example by TCGA and similar projects, the ability to analyze intra-tumor differences emerged only recently and has largely been performed at a limited scale (i.e. across few tumors). Here we sought to perform a systematic analysis of intra-tumor differences through curation of scRNA-seq datasets. We therefore analyzed relative cell states within tumors, avoiding the increased complexity of global cell states.”

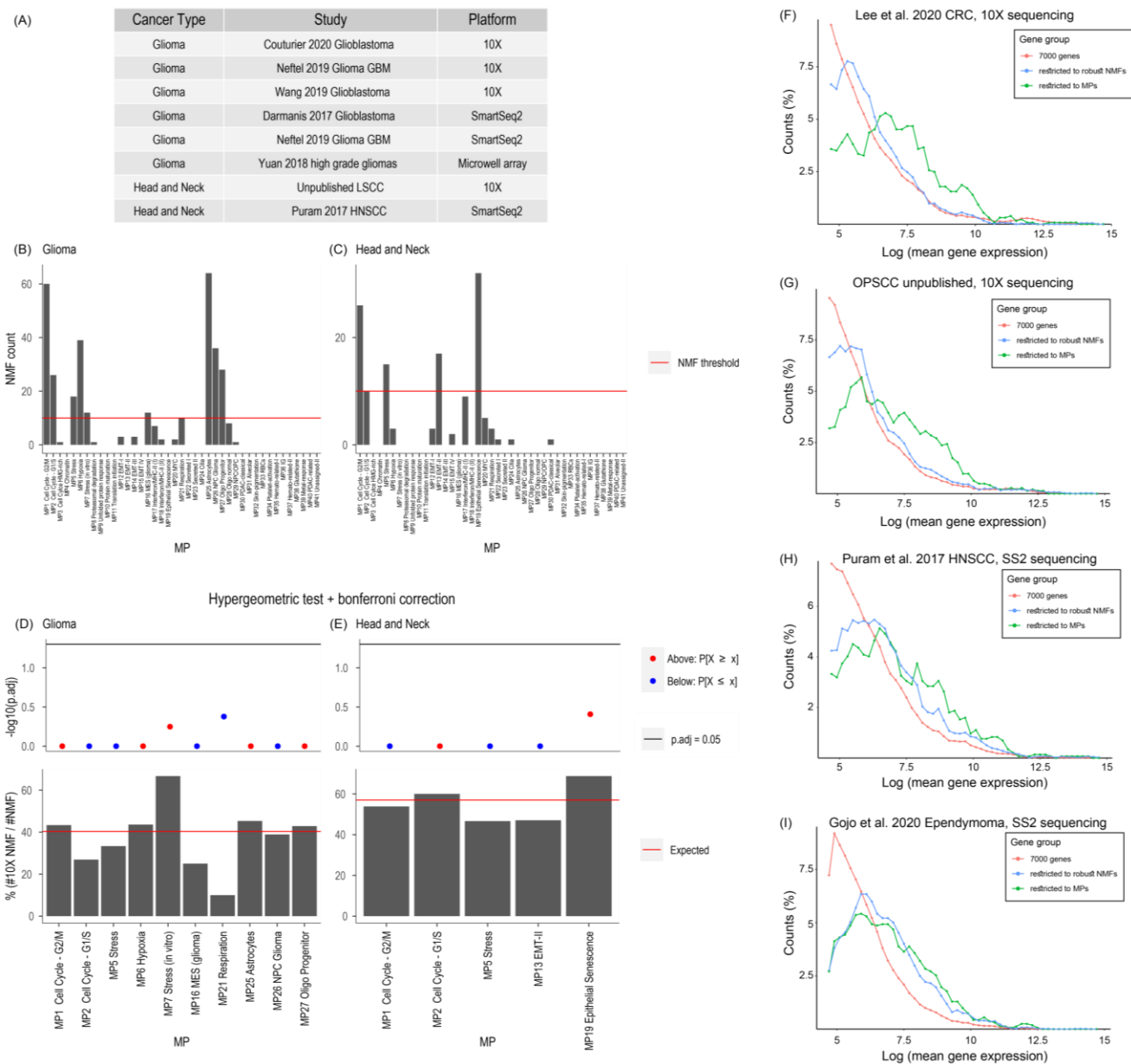
Minor point 2

“Are there any tumor types in the current study simultaneously having full-length data and 10X data? How do different data types influence the analysis?”

To check whether there were biases introduced by integration of different technology platforms, we tested whether MPs were enriched or depleted in a specific platform. We focused on glioma and HNSCC, for which we had samples sequenced with 10X, SmartSeq2 and Microwell platforms (**Response Figure 16A**). Each MP identified in at least 10 samples from these cancer types was evaluated (**Response Figure 16B-C**). For each of those MPs, we compared the fraction of MP-associated NMFs from 10x samples (the most common platform) to the fraction expected based on a uniform distribution across all platforms (**Response Figure 16D-E**). None of the comparisons showed a statistically significant deviation between 10x and the other platforms, suggesting that there isn't a considerable bias between platforms.

Due to other comments, we also examined whether the platform biases the choice of genes for NMF and MP programs to more highly expressed genes in one platform vs. another, and also in this analysis we found comparable patterns between 10x and SmartSeq2 (**Response Figure 16F-I**).

This analysis is included in the manuscript in **page 5** and shown in **Figure S3**.



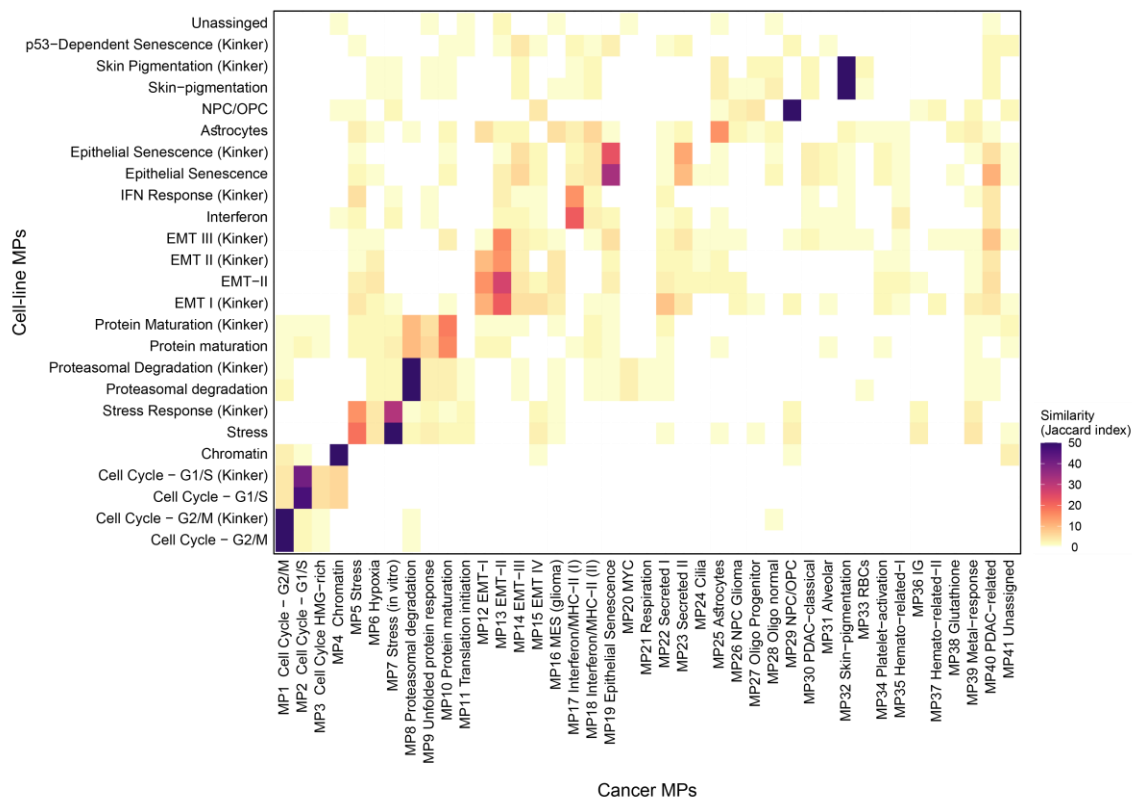
Response Figure 16. Technical controls for potential biases by platform and expression level. (A) Table summarizing the selected studies for this analysis. (B-C) Number of NMF programs derived from glioma and HNSCC samples for each MP. MPs with at least 10 NMF programs were selected (above the red horizontal line). (D-E) The expected percentage of NMF programs from samples sequenced by 10X is delineated by the red horizontal lines. Upper panels show the significance (red above the expected and blue below). None of the cases were statistically significant. (F-I) Distributions of average expression levels for the top 7000 genes (on which NMF was performed) and for genes selected for NMF programs and MPs. Upper 2 studies were sequenced with 10X, and lower 2 panels with Smart-seq2.

Minor point 3

“The authors compared MPs in the manuscript to those previously identified from cancer cell lines. Can the authors provide a supplementary table to summarize the comparison?”

We compared the 12 MPs that were defined solely in cancer cell-lines (Kinker et al. 2020) to the 41 MPs we defined in this study. Since there were some differences between the studies in the methodology for defining MPs and there were several additional cell-lines in our comprehensive cohort that were not included in the original cell-line paper (such as glioma cell-lines), we also repeated the process of defining MPs in cell-lines using the exact same approach used here (e.g. clustering parameters). **Table S4** summarizes the newly defined cell-line MPs, and **Response Figure 17** shows a comparison of the MPs defined in this work to the cell-line MPs (those defined previously and those that were newly defined). Indeed, there is very high overlap between the cell-line MPs and the tumor MPs. Nevertheless, many tumor MPs including common ones (e.g. MYC and respiration) are unique to tumors, and two of the cell-line MPs were not recapitulated in the tumor MPs.

The comparison to cell lines is described in the manuscript in **page 5** and **Table S4**. This Table also includes a comparison to the tumor MPs defined recently by Barkley et al. from a small cohort.



Response Figure 17. Comparison between the cancer MPs and cell-line MPs. Cells-line MPs include those generated previously by Kinker et al. 2020 (and have ‘Kinker’ in parenthesis) and MPs that were newly defined with the exact same method used for the cancer MPs and include some cell-lines that did not appear in the original cell-line study.

Minor point 4

“In the data pre-processing, samples with unresolved CNA patterns were excluded from further analysis. Can the authors clarify how they defined unresolved CNA patterns?”

Cells were defined as having unresolved CNAs and excluded from further analysis in two different scenarios:

1. **Studies for which the classification of cells as malignant vs. non-malignant was not available from the original publication:** In these cases, we applied the methodology that we developed and that was also applied in multiple previous studies in order to classify cells based on inferred CNA profiles. As shown, for example, in **Response Fig. 20F** (see below in page 46), we calculated for each cell two parameters from the inferred CNA profiles: CNA signal (the overall degree of signal for CNAs) and CNA correlation (the consistency of that signal with other malignant cells in the same tumor). Cells that passed predefined thresholds for both of these parameters were defined as malignant, those not passing either one of the thresholds were defined as non-malignant and those cells that passed only one threshold were defined as unresolved.
2. **Studies for which the classification of cells as malignant vs. non-malignant was available from the original publication:** In these cases, our choice was to evaluate if the author’s annotations appear reasonable based on CNAs, but to preferentially retain the published annotations unless we found evidence for misclassifications (in which case we excluded those cells). We chose this conservative strategy because we are not able to deeply interrogate each individual dataset and we assume that in some cases the authors of the original study may have a better understanding of the correct annotations and that it might be difficult to recover such annotations solely from the inferred CNAs. Thus, we first derived the inferred CNA profiles, and then manually inspected those profiles to examine if all clusters of malignant cells are associated with CNAs, while all clusters of non-malignant cells are not associated with CNAs. In cases where a cluster of cells clearly did not show the expected pattern we did not change the annotation but rather excluded the entire cluster from further analysis. In this way, our cell annotations remain consistent with the previous studies but in some cases include fewer cells due to removal of cells with questionable annotations.

Minor point 5

“Can the authors discuss and comment more on MHC II MP? How does the expression of MHC II on tumor cells potentially influence their behaviors?”

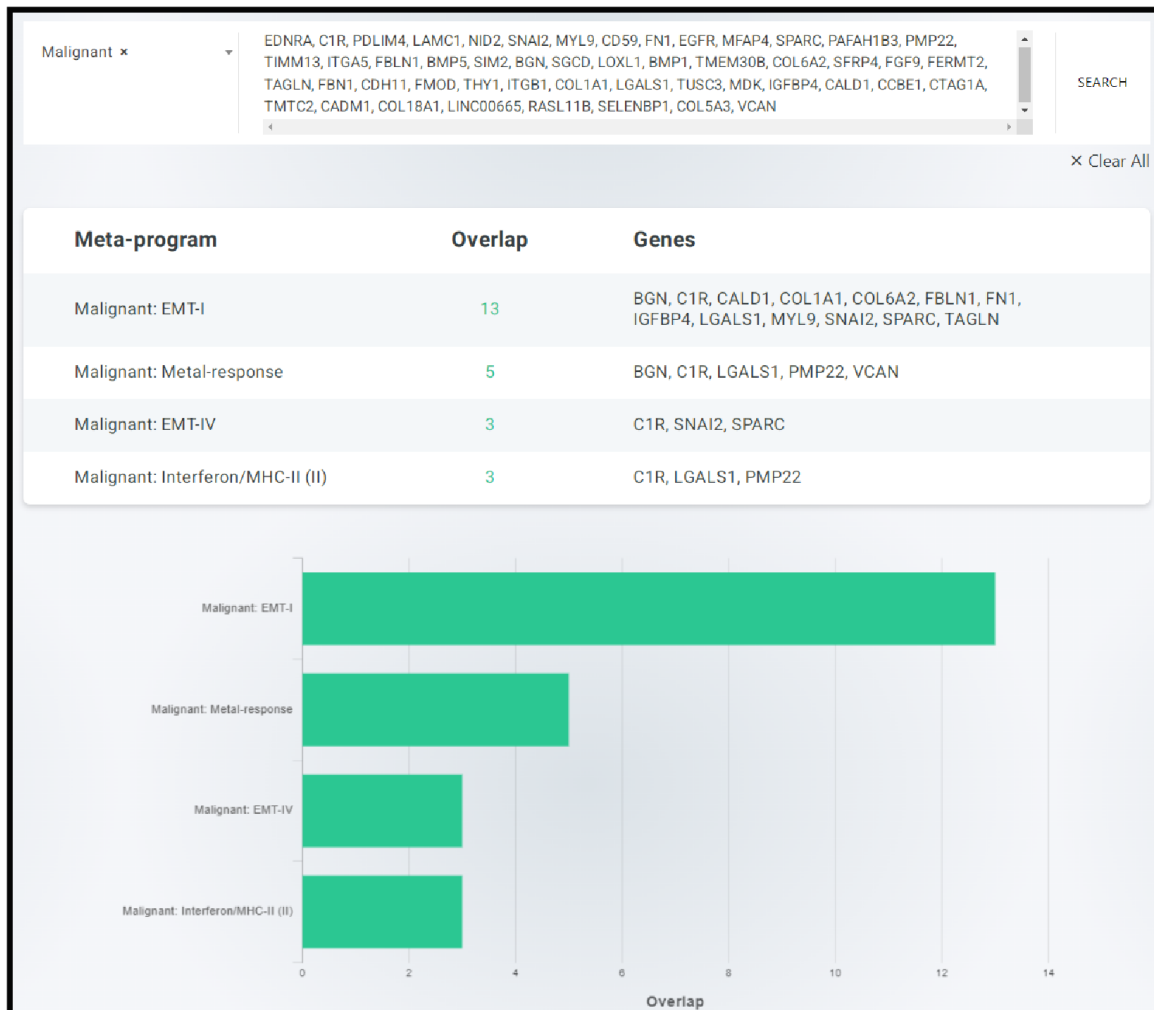
Due to the coupling between interferon response and MHC-II genes that we find in malignant cells (but not in immune and stromal cell types) we expect that interferon signaling is the main cause of MHC-II induction in malignant cells. Such expression of MHC-II may then influence the ability of T-cells to recognize and eliminate the malignant cells, but we currently do not see any evidence for that possibility. We briefly discuss this point in page 11: “Expression of MHC-II and/or interferon responses may have important consequences for the ability of T-cells to recognize and

eliminate malignant cells, raising the possibility that coupling (or decoupling) may be under selection in tumors.”

Minor point 6

“On 3CA website, is there a place that one can query with new sets of genes and compare to the defined MPs?”

We thank the reviewer for this suggestion, as this functionality will improve the user experience on our website and make it easier to incorporate our MPs in their analysis. We have added a new feature within the “Meta-programs” tab of the website that allows the user to input a query gene-set and receive information on its overlap with the defined MPs. This includes a table and bar plots showing the size of overlaps between the query set and the most similar MPs (**Response Figure 18**).



Response Figure 18. Screenshot of the new 3CA page for querying gene sets. Users can input a list of genes and receive information on the overlap of this query set with the MPs from this work.

Reviewer #2

In this study, Gavish et al. collect single-cell RNA-seq datasets from 77 published studies covering 1,163 tumor samples from 24 tumor types to establish an online database. The authors collect published cell type annotations and combine this with de novo annotation and copy-number alteration analysis for calling malignant cells to provide a well-curated and extendable database.

They use this resource to explore intra-tumor heterogeneity (ITH) by inferring meta-programs (MPs) of gene expression that co-vary within malignant cells of specific tumor types or across tumor types. This approach is then also extended to the tumor micro-environment, in order to search for shared meta-programs with malignant cells as well as co-varying meta-programs.

This is a useful resource that will facilitate meta-analyses across large cohorts and different tumor types and provides additional information and curation compared to the original studies. However, the authors do not sufficiently demonstrate that the ITH analysis they perform can lead to genuinely novel and interesting insights. The scope of the MP approach remains limited and many of the observations are not unexpected (such as common MPs between malignant and non-malignant epithelial cells) or analyzed too superficially (see detailed comments). Thus, in its current form I do not see this resource as a major advance for the cancer field.

Comment 1

“The online atlas could be a useful data collection tool for the community. However, I was unable to find the metadata (cell type, sample annotation, and CNA per cell) as a text file that can be used for data analysis. The authors merely provide summary plots. UMAPs highlighting the sample IDs should also be added.”

To address this point and more generally to improve the utility of the 3CA website, we have made an extensive effort to curate as much relevant data (and meta-data) as possible and to provide it in a useful manner through the updated 3CA website. This included:

- (1) Going over all relevant papers to collect all of their available meta-data. In some cases, we also emailed the authors to request for missing annotations.
- (2) Organizing the meta-data in a uniform format to overcome different terminologies and classifications. This was headed by an oncologist in our group to ensure proper interpretation of clinical annotations.
- (3) Updating the website with meta-data files available both in the “Data” link and as a separate “Meta-data” link.

Accordingly, the updated website has been significantly improved, and we will continue to further improve it in the coming months by adding more studies and functionalities.

For each study, the full dataset and annotations can now be obtained using the link in the “Data” column, which downloads a zipped folder. This folder includes the expression data matrix along with CSV files of cell annotations and of sample annotations (i.e. meta-data).

The **cell annotation file** includes cell and sample IDs, cell type annotations including malignant cells (based on CNA inference), and the number of detected genes (“complexity”). Additional cell annotations are available for a subset of studies, such as assignment of cells to CNA-based subclones (following the analysis shown in **Response Figure 2**).

The **sample annotation file** (which is included in the “Data” link but also available as a separate “**Meta-data**” link) includes sample ID, disease name, scRNA-seq platform and number of profiled cells. Importantly, various additional sample annotations are available for subsets of studies. In particular, the following annotations are available for more than half of the studies:

- Patient gender
- Patient age
- Anatomical site
- Histology
- Treated vs. naïve
- Primary vs. metastatic sample
- Extent of disease (local vs. node-positive vs. metastatic)
- New cancer diagnosis vs. recurring disease

Other important annotations are available for a smaller fraction of studies:

- TNM staging
- Grade (well, moderately, poorly or undifferentiated)
- Treatment regimen
- Treatment response
- Clinically-relevant mutations (e.g. BRAF, EGFR)
- Overall survival
- Progression-free survival
- Mutational/hormonal subtype annotations
- Ki67 score
- Smoking status

Except for the downloadable data, we have also significantly updated the pre-computed plots. As suggested by the reviewer, we have now included two UMAP plots per study, one colored by cell types and the other colored by sample IDs. In addition, we provide plots of inferred CNAs, of cell type proportions, of marker gene’s expression, and of the expression pattern of each gene across all cell types and cancer types (see the “Query gene expression” option in the website home page).

Collectively, we are convinced that the website will be a highly important resource and we are committed to continue to improve it and add more data to it (we are already working on curating a second batch of studies that will be added in the future and will be expected to double the size of the compendium).

Comment 2

“The authors report many already known meta-programs associated with tumor cell type variability. There are interesting and important potential programs that were not explicitly reported. In particular, I was curious whether the authors recovered programs associated with immune-suppression, or with the metabolic state of the tumor cells. Those classifications could be important for stratification of tumor types.”

We thank the reviewer for this comment, which highlights two conceptual issues:

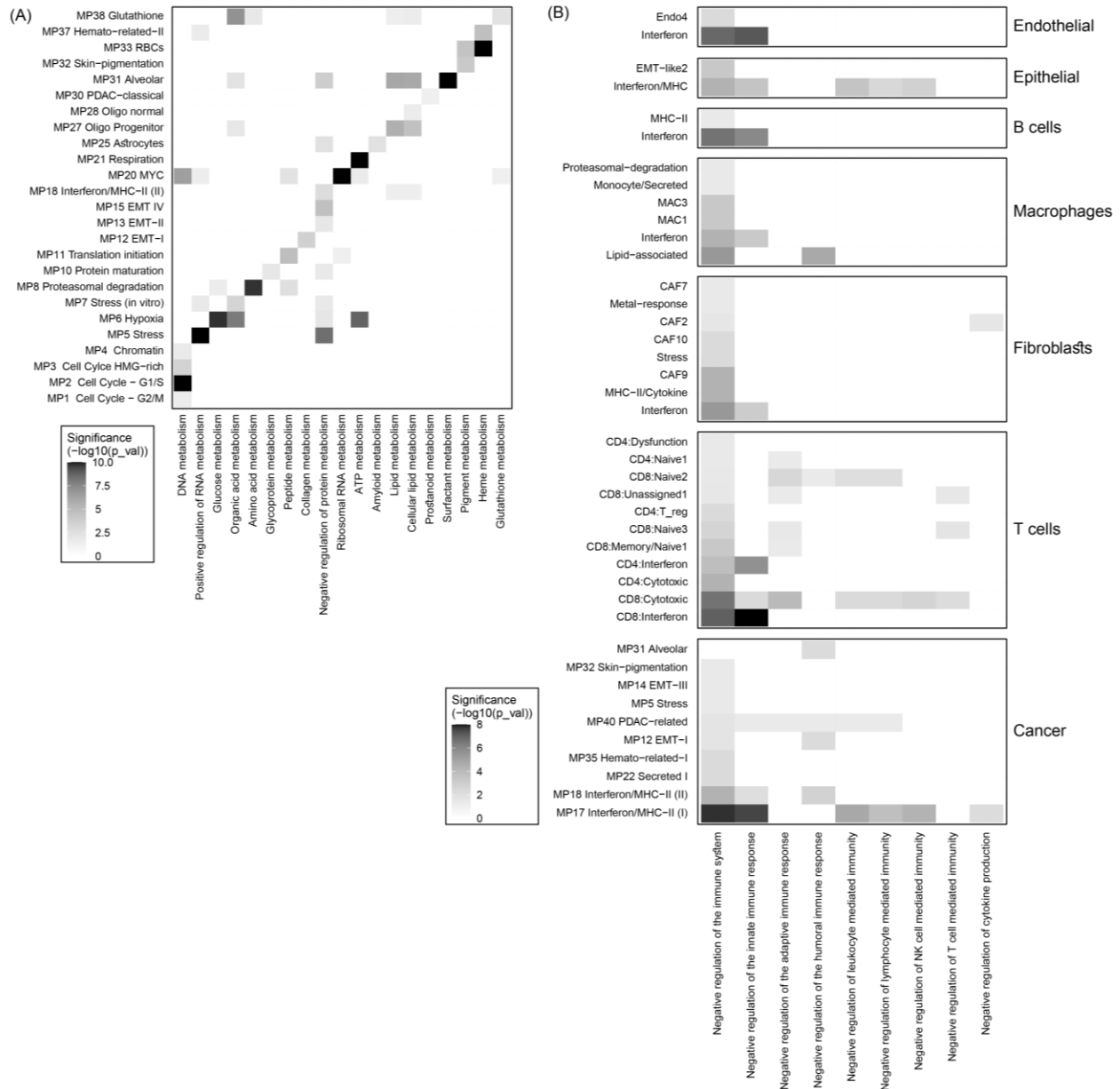
1. Our approach is data-driven, searching for all programs with a strong signal that recurs across multiple samples. Thus, programs that we do not identify are either not recurrent or do not have a sufficiently strong signal (e.g. consist of just handful of genes or associated with limited degree of co-expression). Other studies take a more focused approach, searching for particular genes/pathways of interest and thus would detect such signals even if they are weaker. Yet, our focus is on a systematic exploration of the strongest ITH programs.
2. There is a great deal of ambiguity in how to annotate programs. Certainly, many of the MPs that we describe contain genes associated with immune-suppression or with metabolism and may be defined by some as an “immune-suppressive program” or as a “metabolic program” even though our analysis of functional enrichments (e.g. using Gene Ontology or MsigDB) highlighted other functions and did not lead to such annotations. A simple example of this are the hypoxia programs (in multiple cell types), which are highly enriched with hypoxia-related genes but are also enriched for glycolytic genes, likely reflecting increased glycolysis in the context of hypoxia; thus, we name the program as hypoxia but it could also be named as glycolysis.

We previously defined few MPs in malignant and non-malignant cells that are associated with immunosuppression (e.g. Tregs and T-cell dysfunction) and with metabolism (e.g. B-cell metabolism/MYC, respiration in multiple cell types, lipid-metabolism in fibroblasts and macrophages, glutathione in malignant cells). As noted above, additional MPs might also be associated with these processes.

To further explore this possibility, we compared MPs to functionally annotated gene-sets related to metabolism and to immunosuppression. Metabolism-relates signatures were defined as all gene-sets from MSigDB (including ‘Hallmarks’, curated sets, ‘GO’ annotations, and cell-type specific signatures) that contained in their title the word ‘metabolism’ or words clearly associated with metabolism. Immunosuppression-related signatures were defined similarly based on gene-sets with annotations that included both ‘immune’ and ‘negative regulation’.

We compared the degree of overlap between each of the cancer MPs and the metabolic-related signatures, used a hypergeometric test (including FDR correction for multiple testing) and ranked the significant signatures ($p < 0.05$) for each MP. In **Response Figure 19A**, we show the top metabolic signatures based on their p-value, while also prioritizing signatures with well-defined functionality. Indeed, for a large percentage of our cancer MPs we found significant metabolic related signatures that implied similar functionality to that defined by our MP nomenclature. The full lists of significant signatures is provided in **Table S5**.

A similar approach was applied for the negative immune signatures, where we included MPs from both cancer and TME cell types. The MPs listed in **Response Figure 19B** had at least one associated immunosuppressive signature that passed the significance threshold. This latter analysis, however, is less reliable in our opinion. For example, the T-regulatory program that is well known to be immune-suppressive was actually more enriched in positive regulation than negative regulation of immune responses (data not shown). This reflects a difficulty in confidently assigning genes to immune activation vs. suppression that is often seen in such annotation. Hence, we decided to only include the metabolic-related analysis in the manuscript (**page 5** and **Table S5**).



Response Figure 19. Comparison between MP annotations and signatures related to metabolism or negative immune regulation. (A) Cancer MPs compared to metabolic signatures. (B) Cancer and TME MPs compared to negative immune regulation.

Comment 3

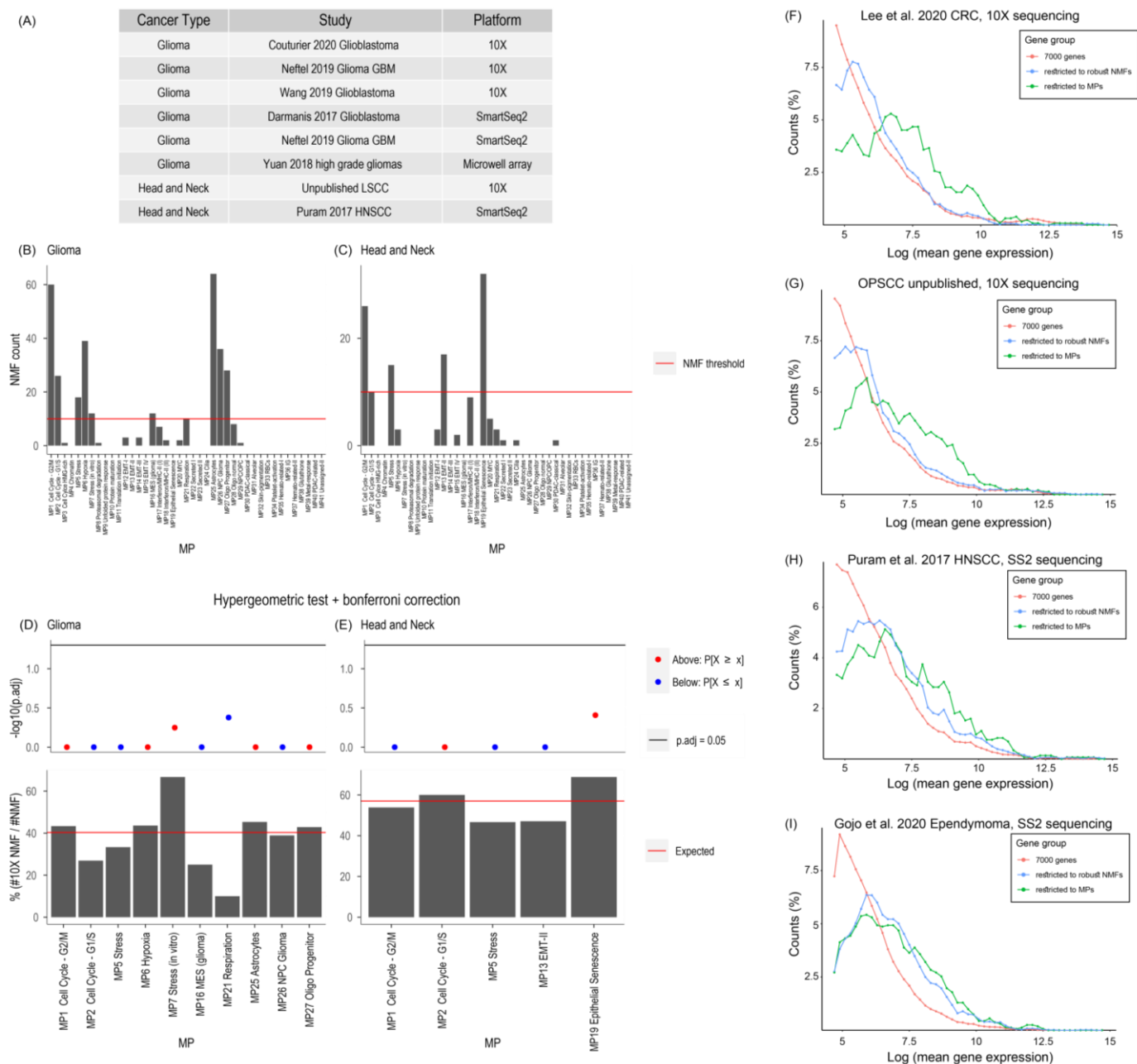
“The Jaccard similarity computed for the inference of consensus metaprograms is sensitive to the input feature set for each sample. If one sample exhibits substantially lower sensitivity compared to another, sets of genes with high Jaccard similarity may be dis-favoured. This could lead to a counterselection of MPs comprising lowly expressed genes, consistent with the finding that the most prominent MPs comprise highly expressed genes (such as proliferation). Did the authors account for global differences in sensitivity across datasets, e.g., by only keeping the intersect of expressed genes across samples?”

We agree with the reviewer that sensitivity and sparsity of single cell data may have considerable influence on such analysis. We are well aware of such issues and have designed the analyses with those in mind. There is indeed a bias against inclusion of very lowly-expressed genes within meta-programs, which is inherent to the data and largely unavoidable. Since lowly expressed genes have less reliable patterns in scRNA-seq data (low signal-to-noise ratio) we believe that such bias is not a problem of our methodology but rather an expected feature of almost any analysis, that in some ways is even desired.

To reduce the influence of such bias on downstream analysis (i.e. to avoid reduced Jaccard indices for programs with many genes that have a low detection rate) it is useful to exclude genes that are only detected in few of the samples. Accordingly, in each study, and for each cell type, we only retain the 7,000 genes with the highest mean expression across all samples in that dataset. This step removes the genes that are only rarely detected. We also argue that this step is helpful as it ensures that our programs are not enriched for lowly expressed genes - in which apparent large differences between cells are often driven by noise (i.e. stochastic sampling).

We note that proliferation genes are overall not highly expressed; in non-cycling cells they are typically not detected as expressed at all, and since the fraction of cycling cells is typically 0-20% (depending on the cell type), their average expression is not higher than most analyzed genes – even among cancer cells.

In **Response Figure16F-I** we show the distributions of mean expression in 4 representative studies that used 10X or Smart-seq2 sequencing, for all 7000 genes (after filtering), for genes from robust NMF programs in each study, and for genes in MPs identified in the corresponding studies. Indeed, we do see some bias towards higher expression levels of MP genes, which we argue is reasonable as the genes with higher expression tend to be more accurate. However, we note that MP genes tend to have only moderately high expression levels and are not among the most highly expressed genes.



Response Figure 16. Technical controls for potential biases by platform and expression level. (A) Table summarizing the selected studies for this analysis. **(B-C)** Number of NMF programs derived from glioma and HNSCC samples for each MP. MPs with at least 10 NMF programs were selected (above the red horizontal line). **(D-E)** The expected percentage of NMF programs from samples sequenced by 10X is delineated by the red horizontal lines. Upper panels show the significance (red above the expected and blue below). None of the cases were statistically significant. **(F-I)** Distributions of average expression levels for the top 7000 genes (on which NMF was performed) and for genes selected for NMF programs and MPs. Upper 2 studies were sequenced with 10X, and lower 2 panels with Smart-seq2.

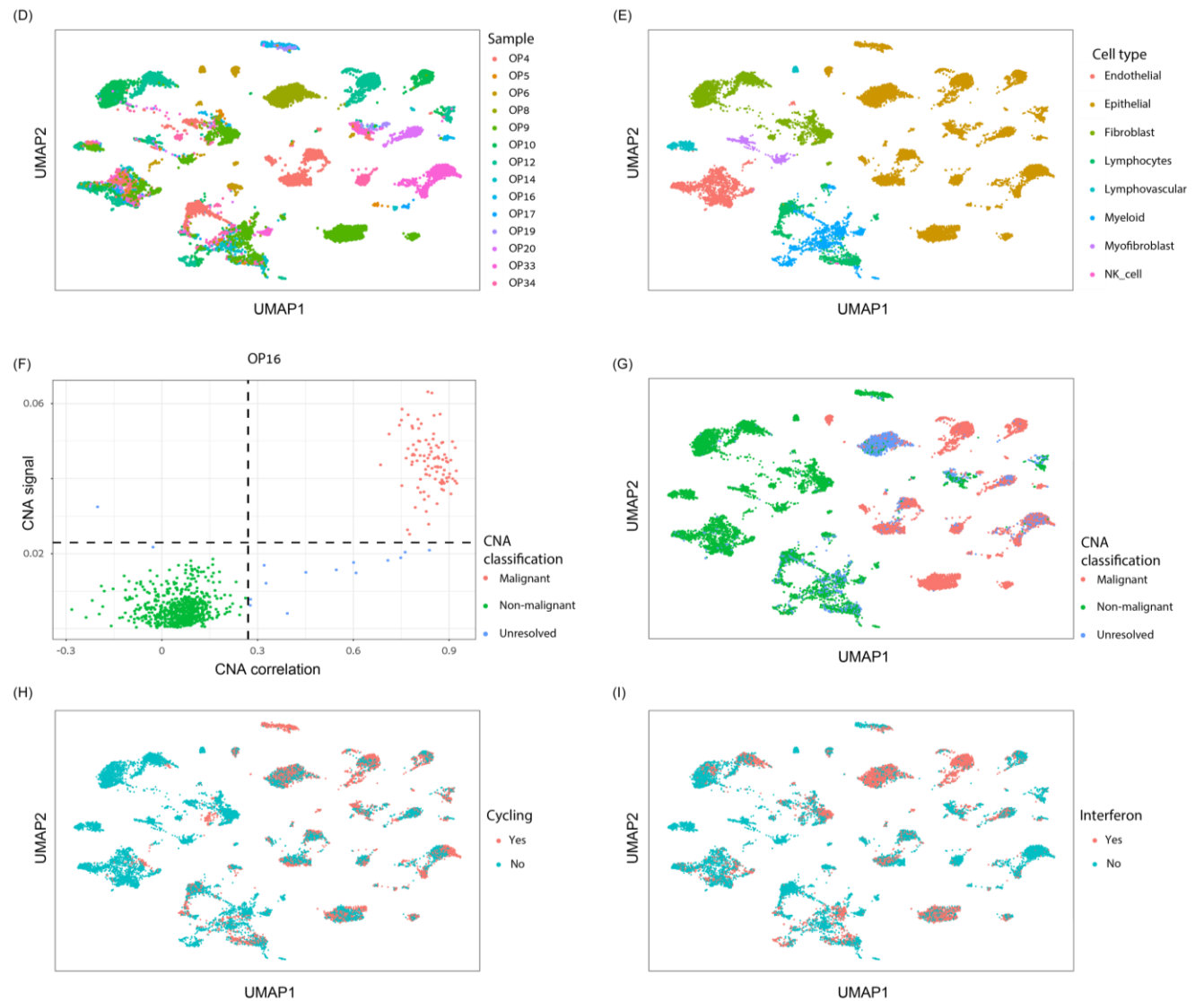
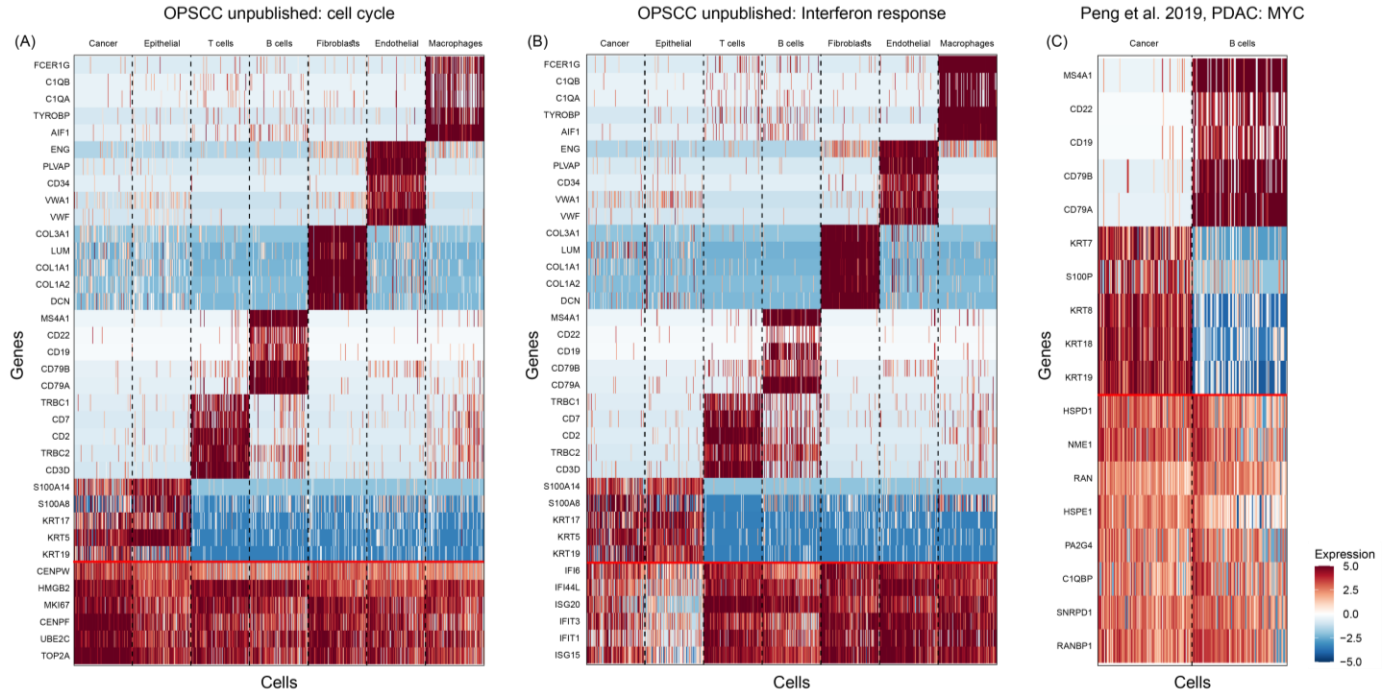
Comment 4

“Please show to what extent cell type clusters correspond to absence or presence of CNA calls. This would show if clusters are uniquely assigned to malignant and non-malignant cells and could address the question whether overlapping meta-programs are due to incorrect annotations of a fraction of cells within the same cluster, e.g., due to sparse coverage or other technical noise. In particular, could the correlation of MYC expression and proliferation MPs between malignant and non-malignant epithelial TME be explained by incorrect annotations of malignant and non-malignant cells?”

To address the concern of incorrect cell type annotations among cells of the overlapping MPs, we first show the expression of cell type markers in those cells from exemplary datasets (**Response figure 20A-C**). We selected 3 of the overlapping MPs – cell cycle, Interferon response and MYC – and plotted the expression of marker genes for the relevant cell types and for the selected MP across all cells with a score above 1 for the corresponding MP (**Response figure 20A-C**). Genes of the overlapping MPs (below the red horizontal line), were indeed highly co-expressed across all cell-types. In contrast, the cell-type marker genes (above the red horizontal line), were specific to the respective cell types. These plots demonstrate that overlapping MPs are indeed expressed across multiple cell types and do not reflect misclassifications.

To further address this point we concentrate on the oropharyngeal SCC (OPSCC) dataset as an example (see UMAP colored by sample and cell type in **Response Figure 20D-E**, respectively), although we note that similar patterns are observed in other datasets. We selected this dataset since it harbored overlapping MPs (cell cycle and interferon response) across multiple different cell types. Several arguments support the expression of overlapping MPs in different cell types and suggest that this is not due to misclassification of cells:

- Our strategy involves assignment of cells to three classes based on their CNA signal and correlations – malignant, non-malignant, and *unresolved cells* that are excluded from further analysis (**Response Figure 20F**). This approach comes at the expense of excluding many unresolved cells but it considerably increases our confidence in the remaining assignments into malignant and non-malignant cells.
- As expected, each cluster harbors cells of mostly one CNA class, and deviations from that primarily include the unresolved cells which are excluded from further analysis (**Response Figure 20G**).
- Overlapping MPs are found in a large number of cells and not in specific outliers as might be expected if this was due to miss-assignments (**Response Figure 20H-I**).



Response Figure 20. Co-expression of MPs is not due to cell miss-annotations. (A-C) Co-expression of selected MP genes in different cell types. Genes above the red horizontal line are cell-type specific, and genes below the red horizontal line belong to the MP. **(D-E)** UMAPs for the OPSCC study annotated by sample of origin or cell type. **(F)** Annotating the malignant epithelial cells was done per sample by gating cells with a high absolute CNA signal and a high correlation across CNA signals. Cells with low values for both were annotated as non-malignant, and cells with intermediate values were unresolved. Representative sample OP16 is shown. **(G)** UMAP according to classification in **F**. A very low degree of miss-classification was observed (less than 5 cells). **(H-I)** UMAPs annotated by whether they scored high (above 1) to the G2/M and Interferon/MHC-II(I) MPs, respectively. Different cell types are seen to co-express these MPs.

Comment 5

“The authors consider the fraction of tumours of a given cancer type for which a robust NMF is assigned to a particular MP as a proxy for the variability of this MP. This interpretation is misleading, since robustness of NMF indicates significant covariation of a group of genes contributing to a factor without implying high variability of a factor per se. If the authors want to make this point they should explicitly show this by a suitable method, such as PCA or fano factor.”

We initially considered several measures for the degree of MP variability within a cancer type. By our judgement, the fraction of tumors with a robust NMF appeared to be a good measure. Nevertheless, we agree that this is difficult to interpret, as there are multiple dimensions of MP variability. Even within a single sample, we could evaluate the fraction of cells expressing the MP, the strength of MP expression in those cells, the consistency between the generic MP and the exact set of co-expressed genes in the particular sample, etc. Thus, each measure has its own properties and advantages/disadvantages. Given this complexity, the comment by this and other reviewers, as well as our general assessment of this section as less informative than others, we have now opted to remove it from the revised manuscript.

Comment 6

“The authors highlight shared MPs between malignant and non-malignant cells. They make some surprising observations such as shared cell cycle and MYC MPs. It is unclear, however, what that really means. Do the authors want to make the claim that MYC or proliferation is variable in non-malignant cells or indeed up-regulated. I am not aware of induction of proliferation in non-malignant cells by tumor cells - perhaps with the exception of immune cell activation - but the data presented in Figure 5 suggest that this is the case for all populations. This should be analyzed in more detail. Would a similar finding be obtained by differential gene expression analysis? The authors should also validate this in situ by immune fluorescence analysis of proliferation or MYC genes together with cancer cell markers. This would strongly support the usefulness of the atlas and the analysis strategy.”

Proliferation

We would like to first clarify the claims regarding proliferation:

- Any cell type in the TME has some (albeit small) fraction of cycling cells. Those cells that are cycling at the time of tumor profiling express the cell cycle MP of the respective cell type.
- The fraction of such cells (for a given cell type) is higher in some tumors than in others. These fractions tend to be correlated between different cell types, suggesting that certain tumors have a trend of high proliferation.
- Such correlations are significantly high only for particular pairs of cell types (edges in Fig. 5A) such as B cells vs. T cells and endothelial cells vs. fibroblasts, and are lower for most other pairs of cell types.

To validate these specific cell cycle associations we have now analyzed extensive spatial proteomic (CODEX) data from three distinct studies of HNSCC and colorectal cancer. Those studies included both a proliferation marker (Ki67) and cell type markers that together enabled us to validate the proposed associations (see below in **Response Fig. 10**). Moreover, we do observe in this CODEX analysis that in certain tumor areas there is an overall tendency of many cell types to proliferate, supporting our prediction.

Interferon responses

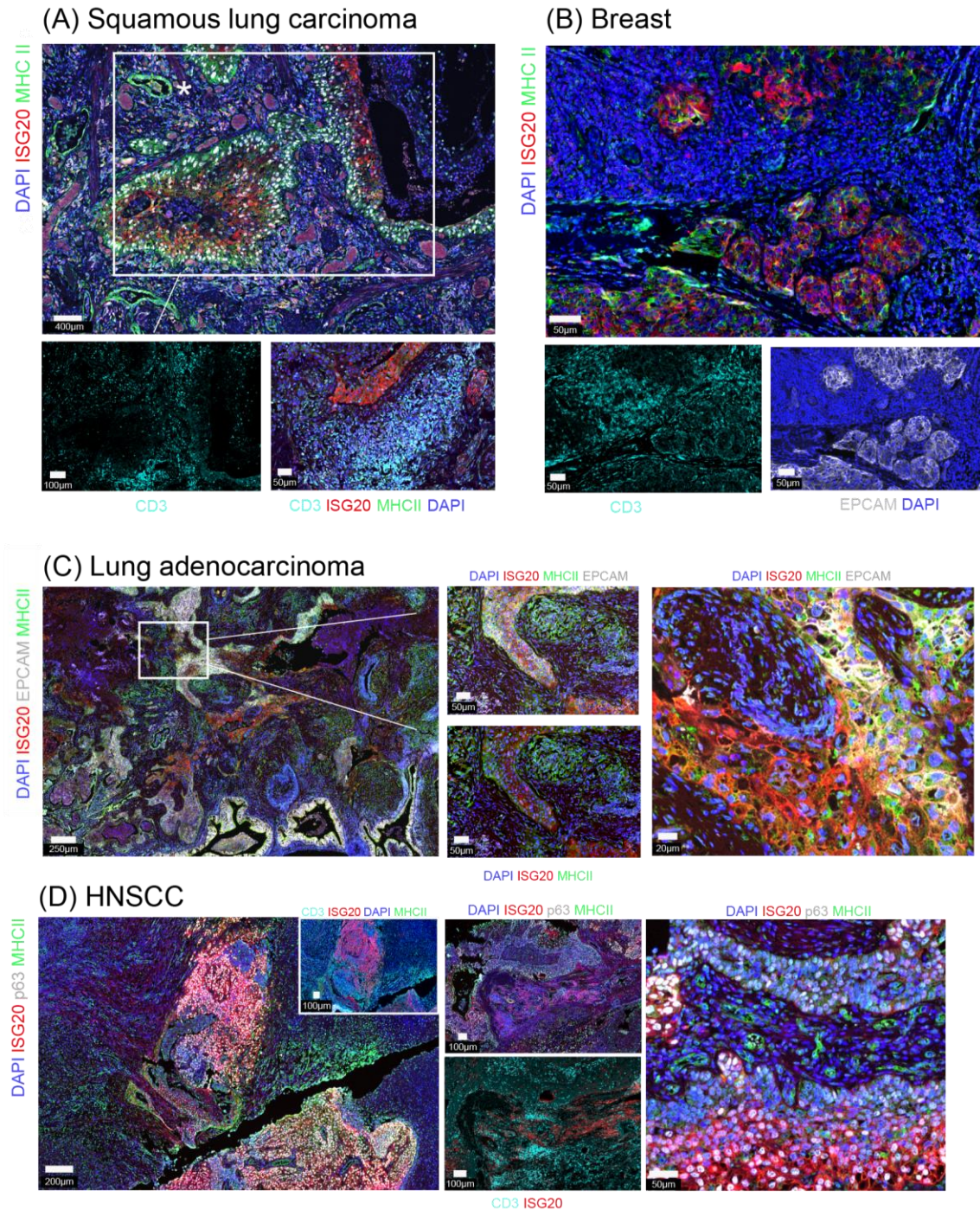
The identification of interferon responses across multiple cell types is supported by analysis of cell type markers (**Response Figure 20** - see comment 4), and was further validated by IHC analysis (see below in **Response Figure 8**).

MYC

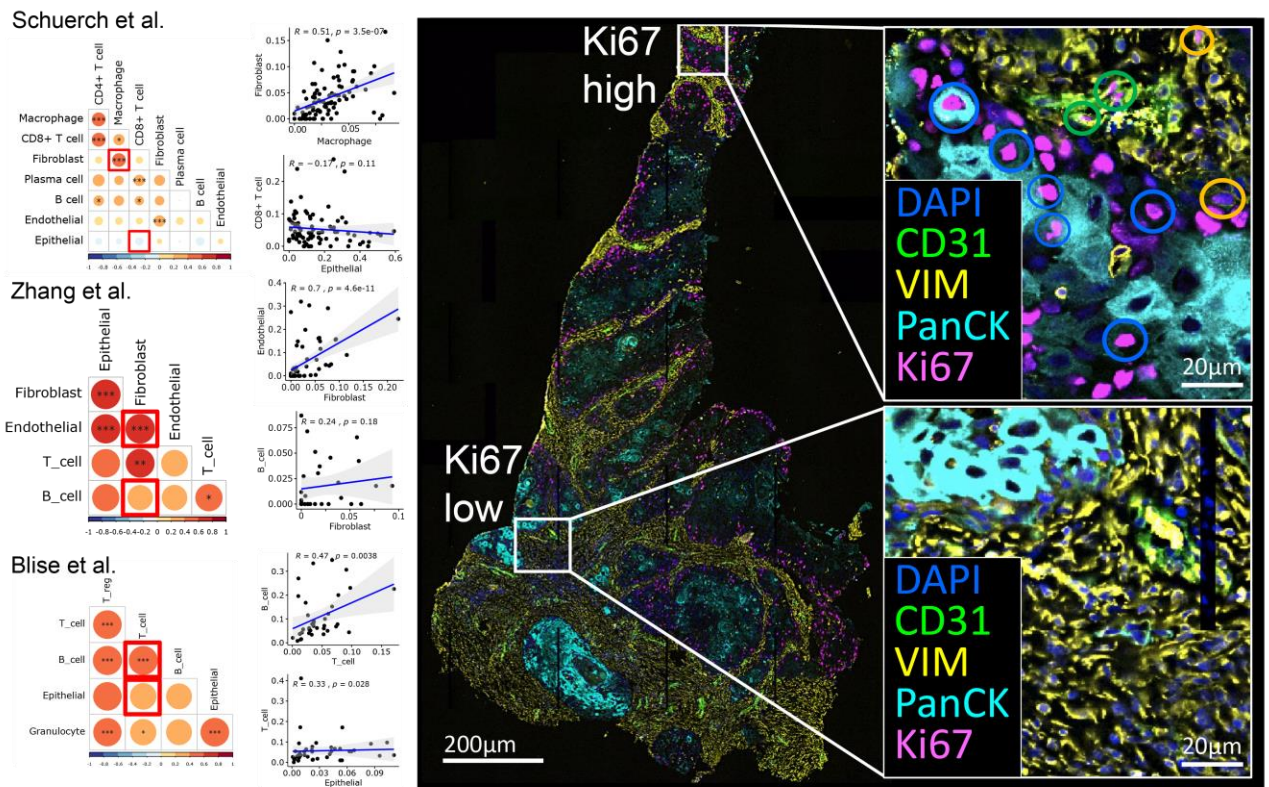
We only see a correlation between MYC-related programs of malignant cells and B cells. We note that the B-cell program is named “metabolism/MYC” because its association with MYC is weaker than that of the malignant cells: it is enriched with MYC targets and has similarity to the malignant MYC program but it does not contain the MYC gene itself and its mechanism of activation remains unclear. This could reflect MYC activation in B cells following an interaction with malignant cells (e.g. through exosomes or nanotubes) but could also reflect some other process that converged on the metabolism/MYC program through an unknown mechanism. We were technically unable to experimentally validate this association. Due to the restriction to B-cells (compared to other shared MPs that are seen in many cell types), the limited connection to MYC itself and the lack of validation, we now soften the claim referring to this observation and significantly reduced the emphasis on this result. We now highlight only the five clusters of MPs with >3 cell types and only briefly mention the MYC interaction (page 12):

“MP clusters were each comprised of MPs that reflect the same process in distinct cell types (blue shade). These include a cluster of interferon responses (in seven cell types), stress responses (in five cell types), heat-shock responses (in four cell types), and cell cycle (in five cell types). The correlations within these MP clusters remained significant when we excluded shared genes between the cell types, indicating that they cannot be explained by ambient RNA (**Fig. S22B**).

These MP clusters likely reflect a concomitant response of multiple adjacent cell types to the same TME features (e.g. interferon and other extracellular molecules). However, the cell cycle cluster appeared surprising given that cell cycle of malignant cells is typically thought to be intrinsically driven by oncogenic mutations. Using several spatial proteomics datasets³⁸⁻⁴⁰ we validated the prediction that cycling cells of multiple cell types are spatially correlated and tend to reside in specific “high-proliferation” niches (**Fig. 5B**, **Fig. S23** and **Fig. S9E-F**). Thus, cell cycle of multiple cell types may be driven by TME mitogens or by direct or indirect interactions between cell types, such as transfer of proteins through exosomes or other mechanisms. Such transfer could potentially contribute also to the other MP clusters described above, as well as to the correlation that we observe between the MYC-related MPs of malignant cells and B cells (**Fig. 5A**), consistent with previous studies⁴¹.”



Response Figure 8. Coupling and decoupling of MHC-II and Interferon response. In malignant cells, MHCII and interferon response are often co-expressed, unlike in non-malignant cells where they are uncoupled (e.g. asterisks in A, right panel in D). Cancer cells co-expressing MHC-II and interferon response frequently form an expression gradient in which the cancer cells co-express both markers on a continuum (upper panels in A and B; panel C; left panel in D). T cells often cluster around cancer cells with high expression of interferon response and MHC class II (left lower panels in A; left lower panel in B; left and middle lower panels in D).



Response Figure 10. Experimental validation using CODEX data. Left panels show quantification of the correlation between cycling cells of different types across studies, together with representative examples (scatter plots). Right panel is an immunofluorescence image of a HNSCC section, highlighting a Ki67-low region (bottom) and a Ki67-high region (top). Right panels: Top box depicts cycling cells (Ki67-high niche) of different cell types that are in spatial proximity, compared to non-cycling cells in the bottom box (Ki67-low niche). PanCK (blue), CD31 (green) and VIM (yellow) mark malignant, endothelial and fibroblast cells, respectively.

Comment 7

“The MP analysis reveals some interesting findings, but remains superficial. For instance, the authors report a correlation of proliferation of CD4+ and CD8+ T cells, but not their abundance, with macrophage abundance. How could this be explained? Naïvely, one would assume that increased proliferation leads to higher abundance, unless T cell are actively killed or migrate out of the tumor tissue. Is there evidence for any of these processes in the MP analysis?”

We thank the reviewer for highlighting this point and agree that these interactions of *macrophage abundance* are difficult to understand. Our focus in the interaction analysis has been on relative frequency of *cellular states* (i.e. MPs) but we also included six features of *cell type abundance*. As described below, we have doubts regarding the *cell type abundance* features and the reviewer’s comment has helped to highlight and increase these doubts. We therefore opted to exclude them

from our revised analysis and include only the features of *cellular states* in which we have much more confidence.

Our doubts stem from two aspects:

1. The only interactions that were found for *cell type abundance* features were the ones mentioned by the reviewer – macrophage abundance correlated with CD4+ and CD8+ T-cell proliferation – which seemed surprising. At the same time, other “expected” interactions of cell type abundances were not identified (e.g. proliferation correlating with abundance of the same cell type, and immune-suppression correlating with macrophage abundance).
2. Across multiple previous studies we saw that (slightly) distinct protocols (and also distinct labs/experimentalists) lead to different cell type proportions and therefore that the bias of scRNA-seq towards/against particular cell types is context-dependent. For example, the overall fraction of T-cells or macrophages varies considerably between studies in a manner that seems difficult to resolve only by cancer type or experimental design. We therefore suspect that such complex dissociation effects hinder the biological relevance of *cell type abundance* features.

Comment 8

“I was surprised that the authors did not observe an association between CD8+ T cell cytotoxicity and malignant cancer cell abundance or proliferation MPs? I would expect a negative correlation. Along the same lines I was wondering whether any MPs related to T cell exhaustion were observed and how they would associate with malignant cell MPs? Is there any correlation between regulatory T cell abundance and malignant MPs? This resource could potentially be very informative for studying immune-tumour crosstalk, but this aspect remained under-investigated.”

We thank the reviewer and agree that the association between T cell states and malignant states is of particular interest. Below we address several aspects related to the reviewer’s questions:

- **Associations with T-cell cytotoxicity:** As noted above, we now removed the *cell type abundance* features from the analysis as we have doubts regarding their biological interpretation. Nevertheless, when focusing on *cellular states*, we do find two associations between CD8+ T-cell cytotoxicity and malignant MPs - the HMG-rich and the proteasomal degradation MPs. The latter association may be linked to the importance of proteasomal degradation for antigen presentation, such that CD8+ cytotoxicity may be correlated to increased antigen presentation by the malignant cells. These interactions are described in the manuscript in **pages 14-15** and shown in **Fig. 5A,C**.
- **T-cell exhaustion:** We observed two T-cell MPs related to exhaustion which we named *CD4+ Dysfunction* and *CD8+ Dysfunction* (see **Table S10**). However, after FDR correction for multiple comparisons (as we did for other analysis) neither of the dysfunction MPs, nor the CD4+ Treg MP, significantly correlated with any of the malignant MPs.

- **Context-specificity:** It is important to note that our *pan-cancer analysis* is aimed at uncovering strong interactions that persist across many contexts, while context-specific interactions are less likely to be detected. For example, in **Fig. S26** we show two interactions of CD4+ T-cells (with malignant and fibroblast MPs) that vary considerably between different contexts and therefore were not identified as significant in the main pan-cancer analysis even though they were significant in specific contexts. As the size of the 3CA cohort grows, we would be better powered to confidently assess context-specific associations, while in this manuscript we focus on the associations that are consistent across many contexts.

Comment 9

“Another important aspect is the capacity of a tumor to metastasize? Do the studies collected here provided information on this aspect? It would be interesting to extract MPs that are specifically associated with the propensity to metastasize.”

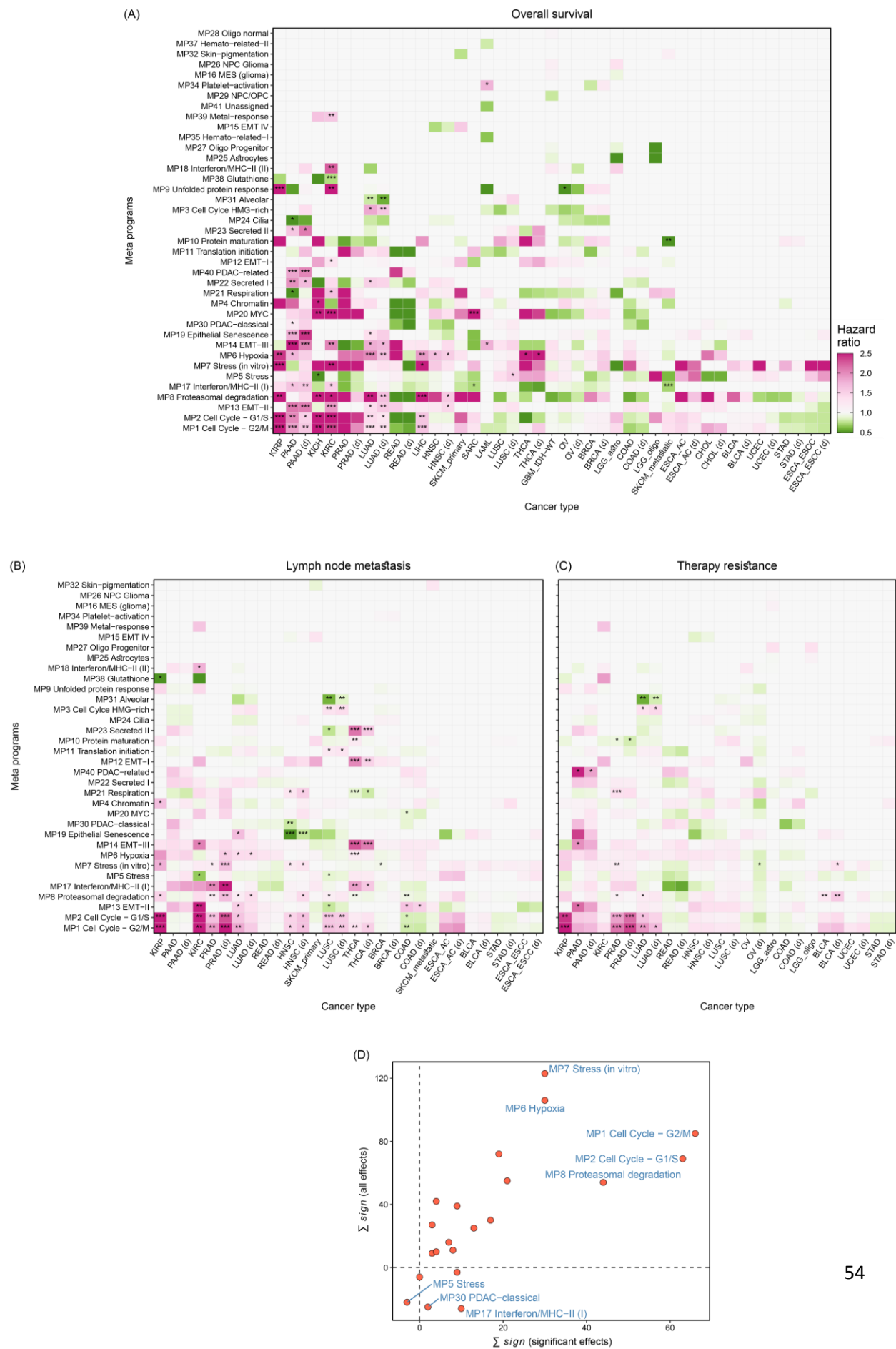
We fully agree with the reviewer on the importance of relating our MPs to metastatic potential, and we discuss this goal below in the context of scRNA-seq and bulk RNA-seq analysis.

scRNA-seq analysis: Unfortunately, there are very few scRNA-seq samples in this cohort from primary tumors of patients which are known to have had metastasis. This reflects both a limitation of clinical annotations but more significantly it represents the clinically reality that metastatic patients are much less likely to have their primary tumors surgically removed. We are therefore currently underpowered to assess the association with metastasis using the single cell data and we will revisit this question as the dataset continues to grow.

Bulk RNA-seq analysis: it is also possible to explore this question by examining expression of the MPs in well-annotated bulk RNA-seq cohorts such as TCGA. To this end, we computed scores for each malignant MP in bulk RNA-seq data for the relevant cancer types in TCGA, and assessed their correlation with various clinical features including lymph node metastasis (**Response Figure 1B**, see below) (we also investigated M stage, but records for this in TCGA are fairly sparse, resulting in limited statistical power).

We do see many significant associations between MPs and the propensity to metastasize, some likely reflecting general features of tumor progression (such as cell cycle) and others more directly suggestive of a specific association with metastasis, such as hypoxia and EMT. It is notable that hypoxia is among the MPs most commonly associated with poor prognosis (**Response Figure 1D**), while EMT is found in almost every cancer type in some form, but there still does not appear to be any MP that is consistently associated with metastasis.

This analysis has been added to the manuscript in **page 7, Figure S8 and Table S8**.



Response Figure 1. Associations of MPs with clinical features in TCGA data. (A) Heatmap showing the hazard ratio of scores for MPs (rows) with respect to overall survival in each cancer type (columns). Purple and green represent association with worse and better survival, respectively. For relevant carcinoma types, the columns of results from data deconvolved by CIBERSORTx are presented with the suffix “(d)”, adjacent to the corresponding column for non-deconvolved data. Significant associations are labelled with ‘*’ for $p < 0.05$, ‘**’ for $p < 0.01$ and ‘***’ for $p < 0.001$. Significance levels were computed via Cox regression and adjusted by Benjamini-Hochberg correction. (B-C) Heatmaps as in (A) showing the difference in average scores between tumors stratified by lymph node metastasis and therapy resistance, respectively. Significance levels were computed by t-test and adjusted by Benjamini-Hochberg correction. (D) Scatterplot showing consistency of associations of MPs with clinical features in TCGA data. Each point represents an MP, its X-value being the sum of the signs (1 for positive effect, -1 for negative effect) of all significant effects (adjusted p value < 0.05) for that MP across all cancer types and clinical features, and its Y-value being the sum of the signs of all effects for that MP. Selected points are labelled with their corresponding MPs.

Reviewer #3

In this manuscript, the authors examine cellular programs underlying intratumor heterogeneity with scRNA-seq data across a large number of studies and over a thousand tumor samples. Their analysis focuses on identifying systematic variable gene expression programs that are conserved across samples. They define 41 consensus “metaprograms” that arise in subpopulations within these tumor samples and provide further classification of these metaprograms into 11 hallmarks of intra tumor heterogeneity. They extend this system to the non-malignant cells in the dataset and use correlations between metaprograms across cell types to infer interactions. Ultimately, this paper outlines a broad description of intratumoral heterogeneity, which I believe will have a pretty significant impact on the field, specifically on how people classify and think about cellular heterogeneity in tumors.

This work builds upon previous work from this group and others describing conserved programs of tumor heterogeneity that exist across different cancer types. The scale of this work is much larger and incorporates a truly impressive number of different data sets. Further they are careful with issues of data integration and batch correction, instead they do not integrate the data, but rather measure the pattern of variability within each tumor.

While the discussion outlines some of the potential reasons that this study may have missed additional hallmarks of intratumoral heterogeneity, I think they likely do capture the largest categories given how many samples these are conserved across. While other studies or groups may attribute different names or classification systems, I generally think that their approach captures most of the variation (at least the more common variation) across tumors.

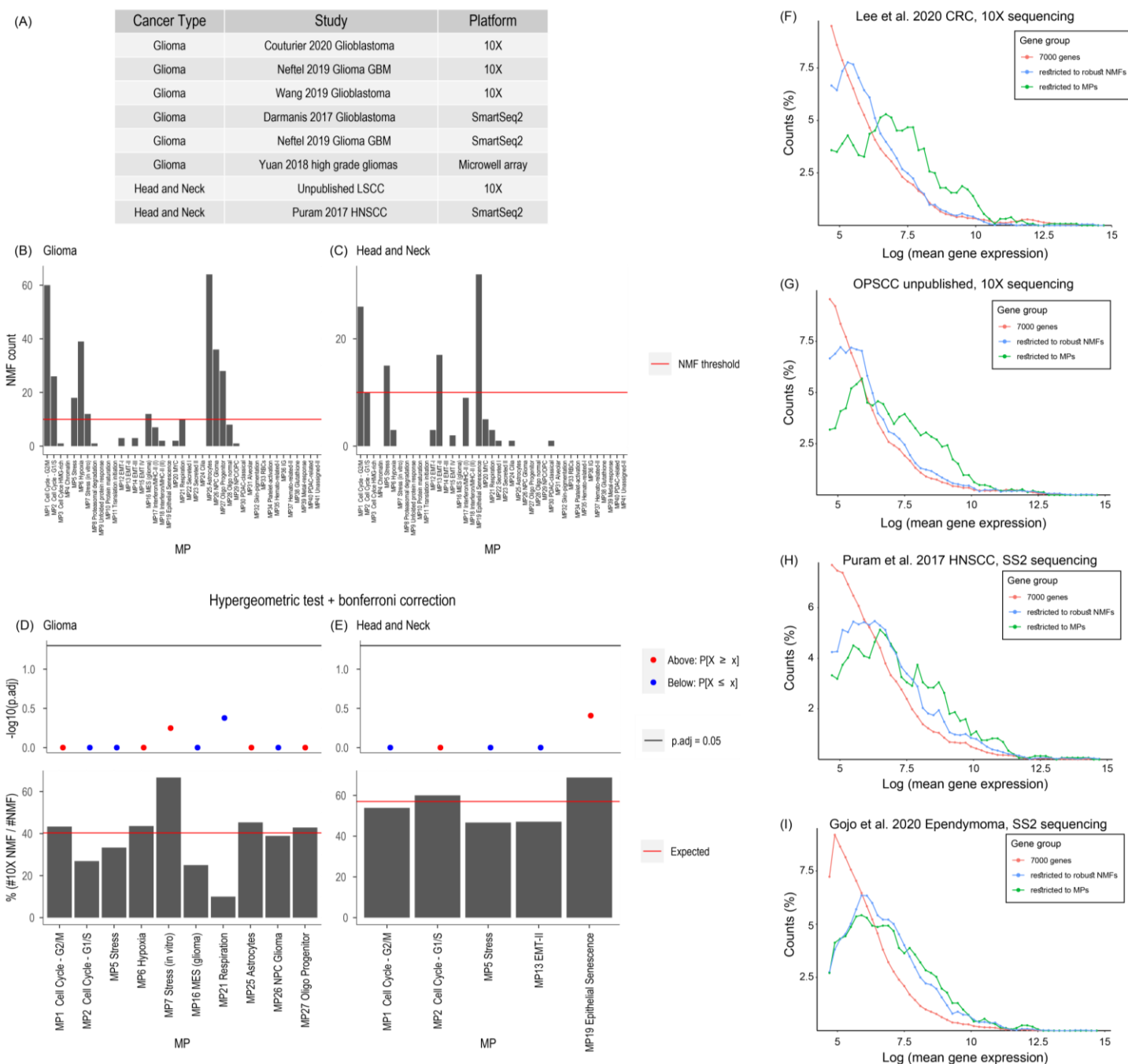
We thank the reviewer for his/her appreciation of our work and its significance.

“In looking at the datasets through the browser, I noticed that many come from a variety of different chemistries. Are there biases or issues arising with the different chemistries used? Are there programs that are only detected from a particular technology platform? Does sequencing chemistry correlate with the ability to recover specific metaprograms? I am wondering if there are considerations related to lowly expressed genes or even genes that are more difficult to sequence for any reason.”

To check whether there were biases introduced by integration of different technology platforms, we tested whether MPs were enriched or depleted in a specific platform. We focused on glioma and HNSCC, for which we had samples sequenced with 10X, SmartSeq2 and Microwell platforms (**Response Figure 16A**, see below). Each MP identified in at least 10 samples from these cancer types was evaluated (**Response Figure 16B-C**). For each of those MPs, we compared the fraction of MP-associated NMFs from 10x samples (the most common platform) to the fraction expected based on a uniform distribution across all platforms (**Response Figure 16D-E**). None of the comparisons showed a statistically significant deviation between 10x and the other platforms, suggesting that there is no considerable bias between platforms.

We also examined whether the platform biases the choice of genes for NMF and MP programs to more highly expressed genes in one platform vs. another, and also in this analysis we found comparable patterns between 10x and SmartSeq2 (**Response Figure 16F-I**).

This analysis is included in the manuscript in **page 5** and shown in **Figure S3**



Response Figure 16. Technical controls for potential biases by platform and expression level. (A) Table summarizing the selected studies for this analysis. **(B-C)** Number of NMF programs derived from glioma and HNSCC samples for each MP. MPs with at least 10 NMF programs were selected (above the red horizontal line). **(D-E)** The expected percentage of NMF programs from samples sequenced by 10X is delineated by the red horizontal lines. Upper panels show the significance (red above the expected and blue below). None of the cases were statistically significant. **(F-I)** Distributions of average expression levels for the top 7000 genes (on which NMF was performed) and for genes selected for NMF programs and MPs. Upper 2 studies were sequenced with 10X, and lower 2 panels with Smart-seq2.

“One question about the web browser, are there compiled datasets available for download or tools to analyze the data across all the samples? Is this intentional and related to the fact that the data is not integrated? From the website, it is clear how to download individual datasets used in this paper, but I think it would be useful to the community to be able to access the compiled data or some form of compiled analysis results (I believe this is what this study really adds in terms of data).”

To enable easier download of multiple datasets, we have added links in the website that download all datasets for a given cancer type. These links have been placed on the bottom row of the page of each cancer-type. We note that the website includes not only the primary published data but also:

- (1) **Cell annotations:** these annotations very often exceed the available annotations of the original studies and hence are of importance to the community. They include CNA-based assignments to malignant vs. non-malignant cells and further assignments of the non-malignant cells to cell types.
- (2) **Sample annotations:** During the revision we have extensively updated the sample annotations (meta-data), including curation of all relevant annotations and application of a uniform annotation format across studies (see response to comment #1 from reviewer #2 in pages 39-40 of this document for a full description)
- (3) **Pre-computed analyses per study:** these include UMAPs, inferred CNAs (including detection of genetic subclones which are included in the cell annotations), cell type composition, and expression pattern of each gene across cell types and cancer types (see “Query gene expression” option in 3CA).

We did not apply integration algorithms to the datasets, since we expect substantial biologically relevant expression differences between tumors, partially arising from their unique cancer types and genetic profiles (per cancer types). We argue that integration methods will inevitably eliminate much of these genuine expression differences, even though they aim to remove only technical effects (for example, see <https://doi.org/10.1101/2021.11.15.468733>). For this reason, our focus has been on analyzing each dataset separately and then combining the *results* rather than the *primary data*. The results of our analyses are available at the website as well as in the supplementary tables.

“How were the cut-offs selected for grouping the different metaprograms into the 11 hallmarks? I see the clustering in 2A, but how were the boundaries defined for including each metaprogram into a hallmark category?”

The borders were defined manually in order to capture well the functional similarities and describe fundamental aspects of cell function or carcinogenesis. We now more clearly acknowledge the subjective nature of that definition in the text (page 5): “MPs were annotated based on their functional enrichments (**Fig. 2C, Table S3**), and MPs with related functions were manually grouped into 11 MP families (**Fig. 2A**)”.

Reviewer Reports on the First Revision:

Referees' comments:

Referee #1 (Remarks to the Author):

I thank the authors for all their replies and for such a herculean effort in adding additional analyses, results, and experimental validations to the manuscript. The manuscript has been substantially revised and improved according to the comments from all three reviewers.

I only have one follow-up question regarding the NMF analysis:

The authors checked whether there were biases introduced by different technology platforms (i.e., 10X vs Smart-seq2) on NMF analysis and showed that there was no considerable bias between platforms. A follow-up question about the NMF analysis: If I understand correctly, in data pre-processing, the authors first normalized UMI counts and centered the data. Then in the NMF step, they set all negative values to zero. It seems by doing so (truncating at zero), the shape of the data distribution would be altered (which is probably quite skewed). I am not sure how exactly this would affect the downstream NMF in terms of identifying gene sets. Have the authors ever compared the NMF results between preprocessing data like this and directly using i.e., raw count data?

Besides, there are a few typos:

Line 34: 'further' rather than 'furthe'? Line 461: 'than' rather than 'then'?

Referee #2 (Remarks to the Author):

In the revised manuscripts the authors have addressed all of my concerns well. In particular, they substantially extended the 3CA online interface to provide relevant primary and processed data, including metadata. This makes this large resource highly valuable for the cancer field.

Moreover, the authors provided clarifications on several aspects and amended their conclusions where necessary.

Importantly, they also included extensive validation to demonstrate that the MP analysis can generate novel insights.

Dominic Grün

Referee #3 (Remarks to the Author):

The authors addressed the comments from my initial review. The improvements to the web browser really help make this an accessible resource to the community.